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ABSTRACT

Credit card interest rates currently average 22%, an 18% spread over the short rate. This spread far exceeds that on any other loan or bond, yet nearly half of households are credit card borrowers. Why are credit card rates so high? To understand this, and the economics of credit card banking, we use regulatory account-level data to analyze the lifetime cash flows of 550 million monthly accounts, representing 90% of the US credit card market. While charge-off rates are comparatively high, averaging around 6%, they explain only a fraction of cards' spread. Reward payments and non-interest expenses are more than offset by interchange and non-interest income. Operating expenses, particularly marketing, are very large, and are used to generate pricing power. Yet, after deducting them, card lending still earns a 6.8% return on assets (ROA), more than four times the banking sector's ROA. Using the cross section of accounts, we estimate that credit card rates price in a 4.3% default risk premium, similar to high-yield bonds. Accounting for this, card lending earns an alpha of around 1.5% relative to the aggregate bank sector.

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1 Introduction

Credit cards are one of the most commonly used financial products in the United States. There are approximately 793 million credit card accounts nationwide, and roughly 78% of US adults have a credit card account in their name.¹ Credit cards are now the primary method of payment for consumers, with total purchase volume of \$6.5 trillion in 2025, or 70% of retail spending.²

Credit cards are also the main form of unsecured consumer borrowing, making them the primary source of marginal buying power for nearly half of households. Indeed, 60% of credit card accounts are borrowers, defined as revolving a balance from one month to the next. Thus, for the majority of cardholders the cost of substituting consumption over time is determined by the interest rate on their credit card, rather than the interest rate on a savings account or Treasury bill, as often assumed in macroeconomic models. This makes a big difference because credit card interest rates are *very* high: the average annual percentage rate (APR) on general purpose credit cards in 2025 was 22%, a spread of 18% over the average federal funds rate.

This interest spread is far higher than that on other loans or bonds. In comparison, in 2025 the spread of rates on Commercial and Industrial (C&I) loans (loans to firms) over the federal funds rate averaged 3.72%, the spread of mortgage rates over Treasuries was 2%, and the spread of high-yield US corporate bonds—the riskiest corporate bonds—over Treasuries was 2.84%.

This leads us to the two questions we ask in this paper. First, why are credit card rates so high? We analyze credit card lending as an asset class and compare its pricing to that of other types of lending. Credit card banking is an intensely retail business, a potentially important factor in understanding credit card rates. Thus, our second question asks what are the economics of the business of credit card banking. We therefore investigate all the streams of revenues and costs involved in this business, and whether market power plays an important role in them. This helps to provide a fuller answer to our first question.

Despite their high interest rates, consumers borrow substantial amounts on credit cards.

¹See [Consumer Financial Protection Bureau – Consumer Credit Card Market Report \(December 2025\)](#).

²We obtain total US credit card purchase volume from the Nilson Report (Issue 1301, February 2026). Total US retail sales and food services spending were obtained from [United States Census Bureau – Advance Monthly Sales for Retail and Food Services](#).

As of the end of 2025, outstanding credit card balances were \$1.1 trillion. Of this, 85% was due to borrowers, with the rest due to “transactors”, accounts that repay their balance by the end of the statement period. Although credit card balances are economically sizable, they accounted for only a modest 4.5% of banks’ balance sheets during the period 2010 to 2025. Yet, because of their high interest rates they generated an average of 16.3% of banks’ annual interest income during the same period.

Because credit card lending is personal lending and unsecured, it is banks’ riskiest type of lending. From 2010 to 2025, credit card charge-offs averaged 4.02% of total balances, compared to only 0.53% and 0.49% for business loans and residential mortgages, respectively. As a result, an average of 55% of banks’ annual default losses were due to credit card lending. Moreover, aggregate credit card default losses are comparable in magnitude to total default losses on corporate bonds, despite the fact that corporate bonds outstanding are ten times as large as credit card balances. Specifically, annual default losses on credit cards were on average 64% of those on corporate bonds during 2010 to 2025, assuming a standard 50% recovery rate on corporate bonds.³

We analyze four potential explanations for why credit card rates are so high. The first is that they are compensation for average default losses. The second is that they are required to cover the high cost of credit card “rewards”, which banks pay their customers in cash or airline miles. The third is that credit card rates price in a large default risk premium, because default risk is undiversifiable and default losses are high in bad economic states. The fourth is that banks are able to charge high rates due to market power.⁴ Market power and risk pricing can also interact if banks price a higher default risk premium into credit card rates than is demanded by investors/lenders in other markets.

Our analysis utilizes a comprehensive supervisory dataset on credit card accounts from the Federal Reserve’s Y-14M reports. Our sample contains monthly data on approximately 550 million individual credit card accounts from the 20 largest banks, covering more than 90% of credit card lending in the US. It contains detailed, account-level information, including the

³Corporate bond default amounts are from [S&P Global Annual U.S. Corporate Default And Rating Transition Study \(2024\)](#).

⁴The literature has argued that the credit card market lacks competition ([Ausubel, 1991](#); [Herkenhoff and Raveendranathan, 2021](#)), has high search or switching costs ([Calem and Mester, 1995](#); [Drozd and Nosal, 2008](#)), and that consumers are heterogeneous in how intensely they search for the best rate ([Stango and Zinman, 2016](#)).

account holder’s FICO score, the account’s APR, credit limit, monthly balance, purchase volume, fees charged, rewards received, and the account holder’s income and geographic location. The data allow us to calculate the various components of revenues and expenses of credit card banking at the individual account level.

We track accounts over time in cohorts based on their month of origination. This is important for calculating the lifetime returns on accounts because a card’s APR spread (the spread of its APR over the federal funds rate) is effectively fixed at origination.⁵ Thus, in setting an account’s APR spread, a bank must consider not only default risk in the near future, but over the account’s whole life. By tracking accounts over time, we capture these future defaults in our measures of the returns banks make on lending to these accounts.

In order to have a sufficiently long time series to capture accounts’ lifetime returns, we focus on the cohorts originated between January 2015 and December 2017, and track them until the end of our sample in December 2023. This gives us at least six years of data for each cohort. We sort the accounts in each cohort into portfolios by their FICO score at origination. FICO is the industry standard credit score, with higher values indicating lower default risk.

We sort accounts by their FICO at origination in order to obtain a large spread in their default risk. This also creates a large spread in their average APR, as FICO is the main variable banks use to set an account’s APR. By tracking the revenues, costs, and defaults of these portfolios over time, we are able to analyze the risks and returns to credit card lending.

In each month, we further sort the accounts in the FICO portfolios into portfolios of borrowers and transactors. This is important because borrowers and transactors generate different risks and revenues. In the months an account is borrowing, it exposes the bank to default risk and is charged interest, unless it is in a zero-interest promotional period.⁶ In the months that an account is a transactor, it poses no default risk and is not charged

⁵This has been the case since the passage of the Credit Card Accountability Responsibility and Disclosure (CARD) Act, which limits banks from increasing APR spread based on existing borrowing. Although banks are allowed to adjust rates with at least 45-day notice, [Nelson \(2025\)](#) shows that in practice it is rare for an account’s APR spread to change during its life. His sample is from 2008 to 2012. We confirm this finding in our extended sample.

⁶Zero-interest promotional periods, typically offered to customers as an incentive to open a new account, normally occur within the account’s first twelve months.

interest.⁷ We find that borrowers make up roughly 60% of active accounts, and that the borrower share of active accounts is strongly decreasing in FICO: for FICO scores above 800 it is slightly below 30%, but it exceeds 65% for all FICO scores below 720, and rises to over 80% for FICO scores near 600.

We begin by analyzing the interest rates that borrowers pay. Two main findings stand out. First, the average APR spread is high for all FICO scores, so even accounts with the highest possible FICO (850) pay an average APR spread of 7.2%. Second, the APR spread is strongly increasing in credit risk, rising to an average of 21% for 600 FICO accounts.

To see if APR spreads are entirely compensation for expected default losses, we compare them to the average net charge-off rates (charge-offs net of recoveries) of the borrower portfolios. Since we track portfolios over a long time, we compute life-time averages of quantities by dividing the portfolio’s cumulative dollar amount of the quantity over the sample period by its cumulative monthly balance over the same period.

We find that charge-off rates decrease almost linearly with FICO, from 9.3% per year at 600 FICO to 1.3% for 850 FICO. Although the charge-off rates are high, especially for the lower FICO scores, they are much lower than the corresponding APR spreads. Thus, the APR spreads are *not* explained solely by compensation for expected default losses. Moreover, the difference between a portfolio’s APR spread and its charge-off rate, which we call its default-adjusted APR spread, is increasing in the portfolio’s default risk: from 6% at 850 FICO to 11.9% at 600 FICO.

Next, we analyze the non-interest income and expenses of credit card banking to see how they affect profitability. Among these, the ones that are unique to credit cards are interchange income and rewards expenses. Interchange fees (also called “swipe” fees) are a percentage of a product’s purchase price (usually around 2%) that is charged to the seller and is split between the credit card network (e.g., Visa, Mastercard, American Express), the merchant’s bank, and the customer’s bank. Most of the fee is earned by the customer’s bank. Rewards are a percentage of the purchase price that is paid to the customer (the cardholder)

⁷An account classified as a transactor in a given month carries no balance beyond the grace period and therefore poses no default risk in that month. If it begins to revolve in a later month, then at that point it is reclassified as a borrower until it pays off its balance (if such a thing occurs) and returns to being a transactor. Thus, classification of accounts into transactors or borrowers is not fixed. This time-varying, account-month classification addresses the concern that a transactor today may become a borrower in the future. It avoids mixing zero-risk and risky account-months, which would blur the relationship between origination FICO, pricing, and realized risk.

by their bank in return for paying with the card. Rewards are paid as cash, airline miles, or points that can be redeemed for cash or travel.

The value of interchange and rewards has become very large. In 2023, the rewards expenses of the six largest card issuers totaled an enormous \$67.9 billion. Thus, it is plausible that banks charge high APR spreads to recoup this expense. However, we find that this is not the case. Rewards expenses are covered by banks' interchange income, with the two being closely intertwined. This is highlighted by the fact that banks' filings report the net of these two quantities as a single category. We find that banks' interchange income is 1.82% of purchase volume on average, while rewards costs are 1.57%.

Net interchange income as a percentage of balances differs greatly between borrowers and transactors. Because transactors repay their balance each month, they have substantially higher purchase volumes, and thus generate correspondingly higher net interchange income. Because they repay their balance, they also have lower balances. We find that net interchange income is about 4.6% of balances for transactors, and only 0.4% for borrowers. As a result, net interchange income accounts for a large fraction of banks' return on assets (ROA) for transactors, but is a minor component for borrowers. Even so, net interchange is positive for both transactors and borrowers for all FICO scores.

We also analyze non-interchange fee income, which includes late fees, annual fees, and balance transfer fees. On average, such fees are around 2.65% of accounts' balances. Fee income is highest at the extreme ends of the FICO distribution: many high-FICO accounts pay large annual fees for premium credit cards, while many low-FICO customers pay substantial fees in order to transfer their balances to new accounts.

Next, we analyze credit cards' operating expenses. Finance research often thinks of banks as portfolios of financial assets and liabilities, and overlooks the influence that non-financial (i.e., operating) expenses have on banks' decisions. A main reason commercial banks have high operating expenses is that they operate a large retail business.⁸ We find that due to their intensely retail orientation, credit card operations have especially large operating expenses: 4-5% of balances annually. These costs explain about half of default-adjusted APR spreads.

A key reason that banks incur high operating costs is to obtain market power, i.e., a

⁸Banks' retail dimension strongly influences their financial characteristics, such as the source and cost of their financing (e.g., deposits) and the assets they invest in ([Drechsler, Savov and Schnabl, 2017, 2021](#)).

“franchise” (Drechsler, Savov and Schnabl, 2021). This appears to be particularly true for credit card operations. We find that one of the largest components of their operating expenses is marketing. Credit card banks spend an average of 1-2% of assets each year on marketing, about 10 times the fraction spent by all other banks. This is why the largest credit card banks, Capital One and American Express, are among the biggest marketers in the world, with marketing budgets as large as those of consumer products giants—and prolific advertisers—like Nike and Coca-Cola.⁹

We use the Y-14 data to analyze the relationship between credit card banks’ operating expenses and pricing power. An advantage of the Y-14 data is that banks report the part of their total operating expenses that is spent on their credit card operations. We find that there is large variation across banks in spending on operating expenses per dollar of credit card balances. Moreover, we show that a bank’s operating expenses per dollar of balances are strongly related to the interest spread it charges borrowers with a given FICO score, and to the gross margin it earns on them. Since different credit cards do not differ much in the service they provide to borrowers, the fact that there are large differences in interest spreads within the same FICO score suggests banks have significant pricing power. Moreover, the relationship between a card’s interest spread and the bank’s operating expenses suggests that operating expenses are incurred in part to attain this pricing power.

We construct a model of credit card lending that formalizes these relationships, which is presented in Appendix B.¹⁰ The model contains two types of borrowers: a group that is highly responsive to marketing, and another group that is not. Some banks optimally choose to pay the fixed cost required to engage in marketing, which attracts marketing-responsive borrowers, thereby giving banks greater pricing power. This allows these banks to charge a higher interest rate markup, which they need to cover their greater fixed costs. As the difference between the two borrower groups is increased, marketing banks increasingly take over the responsive group, and charge a substantially higher rate, while the non-marketing banks take over the non-responsive group by charging a lower rate. Consistent with our analysis, the model shows that the credit card rate can be decomposed into the expected

⁹In 2025, their marketing expenditures were as follows: Nike \$4.7 billion, Coca-Cola \$5.4 billion, Capital One \$5.9 billion, and American Express \$6.3 billion.

¹⁰The model captures the tradeoff that credit card banks face in deciding whether to invest in marketing, and shows how the cost of marketing impacts borrowing rates. Since the model is not required to understand the analysis, we leave it to the Appendix.

loss rate, a default risk premium, and a markup. Under free entry, banks enter until the markup covers their standard operating costs and marketing costs.

Netting all the income and expenses, including operating expenses, we obtain banks' return on assets (ROA) for the credit card portfolios. We find that ROA is high: 6.8% for the aggregate portfolio of borrowers, 2.57% for transactors, and 6.24% overall. Moreover, ROA is strongly increasing in borrowers' default risk, rising from 5% for the high-FICO portfolios to nearly 11% for the low-FICO ones. Thus, ROA's relation to default risk is similar to that of default-adjusted APR. This suggests that credit card rates may price in a risk premium for default risk.

We highlight two findings that further support this hypothesis. The first is that charge-offs are highly correlated across FICO portfolios, rising together in cycles that peak during recessions. Hence, charge-off risk has a common component that cannot be diversified away within the credit card market. Second, credit card charge-off rates are highly correlated with default rates on other types of bank loans and on corporate bonds. Thus, default risk cannot be diversified across loan and bond markets either. Since default risk has this undiversifiable (i.e., systematic) component, it is plausible that this risk is priced.

We test this theory by estimating a single-factor model of default risk on the cross-section of credit card portfolios. As a proxy for the systematic component of default risk, we use changes in the quarterly charge-off rate of the aggregate credit card portfolio. We estimate the beta (exposure) of each FICO portfolio to systematic default risk using a time-series regression of changes in its quarterly charge-off rate on changes in the aggregate charge-off rate. Following the standard asset pricing approach motivated by [Fama and MacBeth \(1973\)](#), we then estimate the compensation for default-risk exposure by regressing portfolios' average ROAs on their estimated betas.¹¹

We find that charge-off betas are strongly and linearly decreasing in accounts' FICO scores. Thus, in addition to predicting the likelihood of default, FICO score is also a good proxy for an account's *beta* to systematic charge-off risk. Regressing portfolios' ROAs on their betas, we estimate the risk premium for charge-off beta to be a highly significant 4.3% per year. Furthermore, we find that the model's fitted ROAs are close to actual ROAs across

¹¹Credit cards are an advantageous setting for analyzing this default risk premium because (a) they have the highest sensitivity of all asset categories to the default cycle, (b) the FICO-sorted portfolios generate a large cross-section of sensitivities, and (c) there are millions of individual accounts, allowing for precise estimates of the FICO portfolios' exposures to the undiversifiable default risk.

the whole range of FICO-sorted portfolios. Thus, exposure to aggregate default risk can fully explain the strong, decreasing relation between FICO score and ROA.

The intercept in the regression of ROA on charge-off beta is the ROA of a hypothetical zero-beta borrower, i.e., one with no systematic default risk. The estimated intercept is 2.66%. We can compare this estimate to the ROA for transactors, which have zero beta by construction, and were *not* included in the regression.¹² The ROA of the transactor portfolio is 2.57%, very similar to the zero-beta estimate.

Next, we analyze how our estimated default risk premium compares to the risk premium in other markets. We focus on the corporate bond market because it is very large and its default rate is also sensitive to the default cycle. Mapping bonds' credit ratings to analogous FICO scores, we find that historical risk premiums for BBB, BB and B rated bonds are very similar to the estimated risk premiums of their corresponding FICO portfolios. Thus, for most of the range of credit/FICO scores, corporate bonds and credit cards appear to price in similar compensation for exposure to default risk. The exceptions are the lowest-rated bonds (CCC/C). In contrast to the linear relation between FICO score and credit card risk premia, our estimated corporate bond risk premiums are comparatively flat for the very lowest credit ratings. As a result, our estimated risk premium on CCC/C-rated bonds is about 3% below that of the correspondingly risky credit cards (620 FICO).

Lastly, we assess whether credit card returns are high relative to the banking sector, even after adjusting for risk. We estimate the "alpha" of credit card lending relative to the banking sector by comparing credit cards' zero-beta rate to the banking sector's ROA adjusted for its small default risk using our risk premium estimate. The banking sector's pre-tax ROA is about 1.5%, and we estimate its default risk premium to be 0.22% to 0.43%. Thus, credit cards' 2.66% zero-beta rate implies an alpha of between 1.38% and 1.59%.¹³

We also conduct a series of robustness exercises to assess the stability of our results and rule out alternative explanations for the high profitability of credit card lending. First, we show that our findings are robust to estimating risk exposures (betas) using realized ROAs

¹²Since borrower/transactor status is reclassified each month, transactors carry no default risk in the periods they are classified as such and thus have zero exposure to default risk, implying a beta of zero by construction.

¹³The charge-off rate of the banking sector is around 0.25% of total assets (0.5% of loans and leases). Using the relationship between the average charge-off rate and default-risk betas estimated from the FICO portfolios implies that the banking sector has a charge-off beta between 0.05 and 0.1, and hence a default-risk premium between 0.22% and 0.43%.

rather than realized charge-offs. Replacing charge-off betas with ROA betas delivers very similar estimates of risk exposures and risk premia across the FICO distribution. This is due to the fact that time variation in realized ROA is driven mainly by time variation in charge-offs, so betas based on ROA or charge-offs capture effectively the same shocks. As a result, the fitted ROAs and implied risk premiums are virtually identical when estimated from ROA betas.

Second, one may be concerned that, because our sample period does not include a major spike in default rates, our estimates of average ROAs, and hence risk premiums, may be biased upward. To address this, we use a less granular but otherwise comparable dataset collected by the Office of the Comptroller of the Currency (OCC) for 2008-2013. This period covers the Global Financial Crisis (GFC), the most severe stress episode for credit card portfolios in recent history.

The OCC data allow us to estimate charge-off betas during the GFC. We find that charge-off betas during the GFC closely match those from our post-2015 sample. However, because the OCC dataset begins during the GFC, it does not allow us to compute lifetime returns for accounts that were originated before the crisis. Instead, we use the Call Reports to estimate the impact of the GFC on average credit card bank-level ROA by extending the sample back to 2008. As expected, credit card banks' ROAs fell substantially during the GFC. Despite this, the average credit card bank ROA is only a little smaller over 2008-2023 than over our 2015-2023 sample. There are two reasons for this. The first is that credit card banks' ROAs are large compared to even the negative shock of the GFC, due to their large default risk premium. The second is that banks managed to earn higher-than-average ROAs in the early post-GFC period, and hence recouped some of their GFC losses. Thus, we find that both risk exposures and average returns are stable across samples, and are not sensitive to the inclusion of the GFC.

Finally, we investigate whether regulatory capital requirements can quantitatively explain the high credit card ROAs and their increasing relationship with borrower risk. We conduct three complementary exercises. First, exploiting differences in risk weights across banks that use the Standardized Approach (SA) vs the Internal Ratings-Based (IRB) approach, we find no evidence that banks facing higher regulatory capital requirements charge higher spreads or earn higher ROAs. Second, we compare credit card ROAs to an upper bound on the cost of regulatory capital, based on the average ROA of non-credit-card banks, and

find that the level and slope of credit card ROAs are far too large to be explained by the cost of regulatory capital. Third, we incorporate regulatory risk weights directly into the risk-premium estimation. We find a small compensation for regulatory capital that only marginally reduces the estimated price of default risk, and has minimal impact on fitted ROAs. Thus, all three exercises indicate that capital regulation cannot explain the level or slope of credit card ROAs.

Related Literature

Our paper contributes to the literature on bank profitability in credit card markets. The academic literature has primarily focused on the effects of the Credit Card Accountability Responsibility and Disclosure Act (CARD Act) on the cost of borrowing and the components of bank profit (Agarwal et al., 2015; Han, Keys and Li, 2018; Nelson, 2025). Agarwal et al. (2015) demonstrate that regulatory limits on credit card fees reduced overall lending income and bank profitability. They also document key components of banks’ realized average revenues, costs, and profits across FICO scores. Their sample covers the pre-CARD Act period which coincided with the Global Financial Crisis. Several recent papers have analyzed the role of reward programs (Agarwal et al., 2022) and the relationship between credit card spreads and default risk (Dempsey and Ionescu, 2025).

We complement this literature in three key ways. First, rather than providing a point-in-time snapshot, as is common in previous studies and industry reports (e.g., Consumer Financial Protection Bureau, 2023), we are one of the first to analyze the lifetime profitability of credit card accounts over multiple years and provide a comprehensive breakdown of the components of revenues, expenses, and profits in the cross-section of origination FICO scores.¹⁴ This approach is important for understanding the pricing of credit card APRs, because they are set at the time of origination and remain largely fixed, due to the restrictions on changing the APR imposed by the CARD Act.

Second, our analysis separates borrower and transactor accounts because they differ fundamentally in the revenues and risks they generate for banks.

Third, we are the first to estimate the default risk premium priced into credit card rates in

¹⁴See Adams, Bord and Katcher (2022) for an overview of the profitability of the credit card business, and see Guttman-Kenney and Shahidinejad (2025), which also highlights the importance of analyzing the lifetime stream of cash flows.

the cross-section of FICO scores. We estimate the exposure (beta) of accounts to the default cycle, the risk premium banks charge for this exposure, and the risk-adjusted average return (alpha) of credit card lending. We also compare the price of systematic default risk embedded in credit card rates with the price of risk for this exposure in the corporate bond market.

Our findings, based on the post-CARD Act period, show that the ROAs on credit card lending are large and decreasing in accounts' FICO scores. Importantly, we find that the ROA on borrowers is much higher than that on transactors, and that this is due to the substantial risk premium they pay for their exposure to the credit default cycle.

More broadly, our paper contributes to the literature on the estimation of risk-adjusted returns in private markets (e.g., [Kaplan and Schoar, 2005](#); [Korteweg and Nagel, 2016](#); [Gupta and Van Nieuwerburgh, 2021](#)). While much of this literature has focused on private equity markets, more recent studies examine private credit markets, including corporate loans ([Flanagan, forthcoming](#)), collateralized loan obligations ([Cordell, Roberts and Schwert, 2023](#)), and private debt funds ([Erel, Flanagan and Weisbach, 2024](#)). We add to this body of work by analyzing credit card loans as an asset class.

Using the Fed's Y-14M reports, the most comprehensive and granular data set available on US credit card accounts, we are the first to estimate the returns, default-risk exposures and risk premia in credit card markets. Our analysis estimates credit card portfolio betas to the default cycle and uses these betas to estimate the default-risk premium. We are also the first to compare credit cards' default risk premium to that of the corporate bond market.

Our findings reveal that credit card rates price in substantial risk premia, ranging from 1-2% for the highest FICO scores to 6.5-9% for the lowest ones. We compare these risk premia to the default-adjusted credit spreads of speculative-grade bonds with comparable default rates and find they are very similar for BBB, BB, and B rated bonds. However, in the case of the riskiest bonds, rated CCC/C, we estimate the risk premium earned by comparably risky credit cards (620 FICO) to be 3% higher. [Fleckenstein and Longstaff \(2022\)](#) estimate the risk premia on credit card asset-backed securities (ABS), bonds backed by credit card cashflows. They estimate significantly lower risk premia, indicating that investors view credit card ABS as relatively safe. However, the risks of the ABS and the underlying credit card accounts cannot be easily compared because these ABS are significantly over-collateralized to make it safe for investors. Issuance of credit card ABS declined dramatically following

the Global Financial Crisis due to regulatory changes.¹⁵

Finally, we contribute to the literature on competitiveness and market power in the credit card market. Prior research has highlighted that this market is characterized by limited competition (Ausubel, 1991; Herkenhoff and Raveendranathan, 2021), high search and switching costs (Calem and Mester, 1995; Berlin and Mester, 2004; Drozd and Nosal, 2008; Galenianos and Gavazza, 2018; Nelson, 2025), significant heterogeneity in consumers’ search intensity for the best rates (Stango and Zinman, 2016), and limited information sharing by incumbent lenders to raise customer acquisition cost (Guttman-Kenney and Shahidinejad, 2025). We contribute to this literature by highlighting a mechanism through which banks attain pricing power in the credit card market: by incurring high operating costs, including large marketing expenses and customer acquisition costs. As we show, banks that spend more on operating expenses are able to charge substantially higher interest rates and earn larger gross margins from borrowers with the same FICO scores. Our findings show that operating costs account for roughly half of cards’ default-adjusted APR spreads.

These results also link our work to studies documenting banks’ substantial market power in deposit markets (e.g., Neumark and Sharpe, 1992; Drechsler, Savov and Schnabl, 2017, 2021). Banks pay significant fixed costs to maintain their deposit franchises, allowing them to charge substantial deposit spreads. Our findings highlight that credit card banks pay large fixed costs to earn substantial spreads on the lending side.

2 Data

2.1 Y-14 Reports

We use a comprehensive and granular supervisory data set on credit card loans from the Federal Reserve’s Y-14M reports. The data are collected for capital assessments and stress

¹⁵The implementation of a change in accounting standards regarding securitizations and special purpose entities (FAS 166 and FAS 167) in 2010 required credit card receivables that were held off-balance-sheet through securitization to be reconsolidated onto the balance sheets of credit card issuers. Due to this, credit card banks significantly reduced their use of securitization and instead held their loans on balance sheet. Industry reports suggest that credit card ABS issuance has declined by 95% between 2007 and 2020 (Structured Finance Association). As of October 2025, only \$85 billion in credit card ABS remained outstanding (IMF US Asset Backed Securities Report (October 24, 2025)), 26.1% of the market’s peak in 2007 (Securities Industry and Financial Markets Association).

tests and contain monthly account-level information. All banks in the US with \$100 billion or more in total consolidated assets are required to report the information. We focus on the reports’ sub-schedule covering credit card loans. These data cover 20 banks, representing more than 90% of credit card lending in the US.

We use account-level variables, including the average percentage rate (APR), FICO score, balance, credit limit, purchase volume, and fees. Banks often provide special benefits to cardholders, such as lower introductory rates, reduced fees, or sign-up bonuses, and we observe these as well. We augment the account-level variables with variables reported as aggregates for portfolios of accounts. Portfolios are defined by the issuing bank, credit card type, and lending type.¹⁶ Portfolio-level variables include interest expenses, rewards, interchange income, operating expenses, and fraud expenses. We use this information to calculate the profitability of the credit card business.

The bulk of our analysis focuses on consumer general-purpose reward credit cards. General purpose cards can be used at all merchants that accept credit cards and account for 90% of all US credit card lending. In contrast, private label cards can only be used at the retailer associated with the card and they make up only a small portion of the data.

We focus on accounts that originated between January 2015 and December 2017, and track them until December 2023, the end of our sample. Although the Y-14 data start in 2012, there are significant gaps in key variables before 2015. By restricting the sample to accounts originated by 2017, we also ensure that the sample is long enough to assess the risk and returns on these accounts over their life cycle. In robustness analysis, we confirm that our findings are not sensitive to the choice of endpoints.

For consistency, we restrict our sample to accounts that meet the following criteria: (1) observations begin within one month of the account’s origination, (2) the balance variable is not missing, and (3) if the account exits the dataset, the reason is specified as either charge-off or closure; otherwise, the account remains active until the end of our sample. In contrast to the usual practice of sampling the data, e.g., studying subsamples of 1% of observations, we analyze the entire data set. This gives us an average of 31 million observations per month for the cohorts we study for the period January 2015 to December 2023.

¹⁶There are four credit card types: general purpose, private label, business card, and corporate card. There are also four lending types: consumer bank card, consumer charge card, non-consumer card, and non-consumer charge card.

2.2 Other Data Sources

Call Reports We complement the Y-14 data with bank call reports, which contain quarterly data on the income statements and balance sheets of all US banks. We use the call reports to compare the profitability of credit card banks to other banks. We identify credit card banks as those in which (1) personal loans’ share of assets exceeds 50%, and (2) at least 90% of personal lending is credit card lending. This is the definition used by the Federal Reserve’s report to Congress on credit card banking.¹⁷

Corporate Bond Data We obtain data on corporate bond issues from the Mergent Fixed Income Securities Database (FISD). We use the data to compare the risks and returns of investing in corporate bonds to those of credit card lending. The data set contains information on a bond’s issuance date, issue size, coupon rate, maturity date, and credit rating.¹⁸ It also provides information that allows us to calculate the return to investing in a given bond issue. In particular, we observe if the bond defaulted (and when), reached maturity and repaid its entire principal, or was called by the issuer.

3 Overview

Table 1 panel A presents summary statistics for key income and expense components in credit card banking, based on data aggregated at the bank-month level for the period January 2015 until December 2023. Credit card banking is characterized by a high interest income rate, with an average interest spread—defined as the interest income rate minus interest expense rate (which we set equal to the federal funds rate) of 11.8%. Charge-off rates are substantially smaller, averaging 5% across banks in the sample, with relatively low variation. Recovery rates are low, averaging just 0.5%, or roughly one-tenth of charge-off rates. Interchange income and rewards expenses, reported as a share of assets to ensure comparability with net interest income, are both significant in magnitude. The average interchange income rate is 7.1% of assets annualized, while rewards expenses amount to 4.8% of assets, making

¹⁷The series of such reports are known as the “Report to the Congress on the Profitability of Credit Card Operations of Depository Institutions”: <https://www.federalreserve.gov/publications/credit-card-profitability.htm>.

¹⁸We use Standard and Poor’s ratings as the primary source of credit ratings.

interchange income the larger of the two. Additionally, banks earn 2.1% of assets per year in non-interest income, with 1.7% coming from various fees. Operational costs represent the largest non-interest expense, averaging 4.9% of assets annually and varying significantly across banks. Other non-interest expense components are relatively small, with fraud-related expenses accounting for just 0.3% of assets per year.

3.1 Borrowers and Transactors

A key distinction among credit card accounts is whether an account is a transactor or a borrower (known as a “revolver” in industry terminology), in a given month. Transactors are accounts that repay their balance in full by the end of the statement month, whereas borrowers carry a balance and borrow the unpaid amount from the bank. This distinction is important because a transactor does not pose a default risk to the bank and pays no interest. In contrast, a borrower exposes the bank to default risk and is charged interest. The exception is if the borrower is in a zero-interest promotional period. Such promotions are offered to prospective customers as an incentive to open a new account or transfer existing borrowing from another bank (“balance transfer”), with the promotional period typically occurring in the account’s first twelve months.

Therefore, in each month, we classify each account as a transactor or borrower. We classify an account as a borrower if the account either repays less than its previous balance, incurs finance (i.e., interest) charges in the following month, experiences a charge-off, or transferred a balance from another account. All other accounts are transactors (they do not borrow). We define a transactor as *active* in a given month if it has either a positive cycle-ending balance, purchase volume, or payment activity in that month. Inactive transactors do not use balance sheet capacity or generate any income for the bank.

Table 1 panel B reports summary statistics computed separately for borrowers and active transactors. A significant minority of accounts are inactive and because they both cost and contribute little, we focus on the active transactors to understand the characteristics of the credit cards that are actually used.

There are several key differences between borrowers and active transactors. Borrowers’ average daily balance (ADB) is much larger on average than transactors’ ADB. This is because borrowers’ ADB includes both their current month’s purchase volume and their

past borrowing. Since transactors do not borrow, their ADB is due only to the current month’s purchase volume. In contrast, active transactors’ purchase volume is larger than borrowers’, in part because they are not encumbered by debt. Since interchange income and rewards payments are proportional to purchase volume, they are larger for transactors.

3.2 Distribution of Credit Scores

To study the role of credit risk in credit card banking, we sort accounts by their FICO score at the time of account origination into 50 evenly spaced bins spanning 600 to 850. We use 600 as the lower cutoff because it is rare for an account with a lower score than this to be originated. FICO score is the industry standard credit score. It is intended to predict an account’s credit risk relative to other accounts (not its absolute level of risk, which depends on aggregate factors). In order to confirm that origination FICO score is a strong predictor of future charge-off rates in the cross-section of accounts, we estimate a series of regressions of charge-off rate on different sets of controls and fixed effects. The results, presented in Appendix [Table D1](#), demonstrate that FICO score has significant predictive power for ex-post default rates. A basic model including only linear FICO terms explains nearly 66% of the variation in charge-off rates. Additionally, a substantial portion of charge-off variation is attributable to bank-specific factors, and accounting for these adds 18 percentage points to the explanatory power. Finally, interacting origination FICO with bank fixed effects further improves the predictive power, adding 5 percentage points and ultimately explaining 91% of the variation in charge-offs.

Because of its informativeness about credit risk, banks use the FICO score to set accounts’ borrowing terms. In particular, an account’s origination FICO score strongly predicts its APR spread, which is set at origination and usually remains fixed for the account’s life. This is due, at least in part, to the CARD Act, which prohibits banks from increasing an account’s APR spread on outstanding borrowing. Appendix [Table D2](#) shows that fixed effects for origination FICO-origination date and origination FICO-bank explain over 96% of the variation in the cross-section of APR spread. These results suggest that the origination FICO explains a large share of the cross-sectional variation in APR spreads.

We begin the analysis by examining the distribution of FICO scores across accounts. Panel (a) of [Figure 1](#) plots the distribution of FICO scores across accounts. It plots the

FICO share of newly originated accounts (blue), all accounts (red), and the ADB-weighted share of all accounts (green). The distributions of new and existing accounts are close to uniform for FICO scores between 670 and 820, with a mode at 695.¹⁹ In contrast, the ADB-weighted share is highest for FICO scores between 660 and 750.

Within each FICO bin, we further sort the accounts into portfolios of borrowers and transactors. This is important because borrowers and transactors have significantly different income and expense streams. Panel (b) of [Figure 1](#) plots the share of borrowers across the FICO bins, on an equal-weighted basis (blue) and weighted by balances (red). The share of borrowers is monotonically decreasing in FICO score. At 600 FICO, over 80% of accounts are borrowers. This drops to 50% for 750 FICO, and decreases to a little less than 30% for FICO scores over 800 (“superprime”). Because borrowers have substantially higher balances than transactors (see Panel (c) of [Figure 1](#)), their balance-weighted share is substantially higher at all FICO scores. It is close to 100% for the lowest FICO and remains around 90% for FICO scores of up to 750. It is above 50% even for FICO scores above 800. Thus, lending to borrowers accounts for most of credit card banks’ balance sheets.

4 Decomposition of Income and Expense Streams

We analyze all major income and expense streams for accounts in the cross section of FICO scores. Importantly, we do so separately for borrowers and transactors, since they differ substantially across the two groups. We begin with the APRs charged by banks.

4.1 APRs and Credit Losses

Panel (a) of [Figure 2](#) plots the effective interest rate (APR) paid by borrowers across FICO scores. We compute the effective APR from the reported interest payments and ADB and subtract the federal funds rate to get the effective APR spread. The effective APR may differ from the reported APR because of promotions, during which a lower interest rate is charged (e.g., 0%). This is most common in the account’s first year, but also occurs at other

¹⁹The new and existing account distributions are nearly identical. This is because attrition rates are very similar for each origination FICO bin. Attrition rates are the share of accounts closed, which can occur due to default, the customer voluntarily closing the account, the card getting lost or stolen, the customer dying, or other reasons.

times. Using the effective APR allows us to account for the impact of promotions on banks' interest income and consumers' cost of credit. We observe that a significant number of borrowers with high FICO scores at origination take advantage of 0% APR offers to revolve credit. Notably, for borrowing accounts with an initial FICO score of 750 or higher, the 25th percentile of effective APR is 0%.

It is important to note that almost all credit card APRs are a constant spread over the prime rate, which is itself a constant 3.0% spread over the federal funds rate.²⁰ Thus, it is natural to analyze APRs as a spread to the federal funds rate, which we call the effective interest (or APR) spread. The values of the FICO bins' APR spreads are reported in [Table 2](#). The APR spreads are large. The average APR spread across FICO scores is 14.5%, and even the smallest APR spread exceeds 7%. These spreads are much larger than those on other common types of bonds or loans. For instance, over 1997-2025 the average credit spread on investment-grade and high-yield corporate bonds was only 1.5% and 5.2%, respectively. Even the most distressed high-yield bonds, those rated CCC and below, had an average credit spread (11.0%) smaller than that of most credit cards.

Since credit card lending is unsecured lending to consumers, it is exposed to a significant risk of credit losses. Hence, we first analyze whether the large APR spreads are explained by similarly large credit losses.

Panel (b) of [Figure 2](#) is a stacked plot comparing the interest income (blue) of borrower accounts in the FICO bins against their credit losses (green) and interest expenses (red). These quantities are plotted on the left axis. Income is plotted as a positive quantity, while losses and expenses are negative. We focus on borrowers, as transactors do not have interest income or credit losses.

The net credit loss rate measures the quantity of charge-offs per dollar of lending per year over our sample. We net charge-offs against any recoveries, which for credit card lending are small. Interest income is the interest paid by the accounts. Hence, the plotted rate is exactly the average effective APR rate in Panel (a) of [Figure 2](#). We assume the interest rate

²⁰As of 2022, 98 percent of general purpose accounts in the Y-14 were variable rate cards according to [Consumer Financial Protection Bureau \(2023\)](#). In a highly-cited paper, [Ausubel \(1991\)](#) documented that during the 1980s credit card APRs had low sensitivity to the federal funds rate. This has not been true for a long time, as credit card APRs have moved closely with the federal funds rate since at least 1995.

paid on financing equals the federal funds rate.²¹ Thus, the difference between the interest income and expense rates equals the effective APR spread reported in Table 2. In analogy to bonds, we can also view it as the credit spread on this credit card lending.

Since our goal is to understand the profitability of accounts over their lifetime, we plot the annual average rates of these quantities from the time of origination until the end of the sample. To calculate the lifetime average rates for each origination FICO bin, we divide the cumulative monthly dollar amount of the quantity of interest (e.g., net charge-offs) for all accounts in the FICO bin over the sample period by their cumulative monthly balances over the same period. This ratio gives the rate of the given quantity per dollar of monthly balance. To annualize it, we multiply it by 12. To combine the cohorts (origination months) of a FICO score bin, we effectively treat them as one large portfolio and sum the numerator (e.g., net charge-offs over the sample) for all the cohorts of the FICO bin and divide by the sum of their balances. We follow this same approach for all the quantities plotted in the figures below.

Panel (b) of Figure 2 confirms that lifetime credit losses are substantial for many FICO scores. Moreover, they are strongly, nearly linearly, decreasing in FICO score. At the lowest origination FICO score (600), lifetime net credit losses are a very high 9.3% of balances per year. By 720 this decreases to a still large 5.7% of balances per year over the lifetime, which is also the loss rate for credit card lending in the aggregate in the sample. Even at 800 origination FICO (“super-prime” according to Experian), the credit loss rate is still 3.2%. By 850 FICO, the highest possible, the credit loss rate drops to 1.3% per year. Note that these FICO scores are the accounts’ scores at the time of *origination*, not at the time of default. By the time an account defaults, its FICO score has usually deteriorated substantially. However, we need to map accounts’ credit losses to their origination FICO scores in order to compare them with their APR spreads, which are fixed at origination. In addition to default-related charge-offs, there are also charge-offs due to fraud. The cost of these is 0.18% of ADB per year, so far smaller than credit-related charge-offs.

²¹There are two main reasons why we believe this is appropriate. First, the interest expense rates of banks that engage mainly in credit card lending, such as American Express, Discover, or Synchrony bank, are very close to the federal funds rate. The reason is that they lack a deposit franchise and hence fund themselves mostly by selling short-term CDs at competitive rates, i.e., close to the federal funds rate. Second, for banks that do have a deposit franchise, and hence obtain deposit financing at lower rates, the difference should be viewed as a payoff to their deposit-taking business, which involves its own substantial costs, not their credit card business. Thus, to be conservative we set the interest expense rates to the Fed Funds rate.

To test whether APR spreads are just compensation for average default losses, we subtract the credit loss rate for each FICO bin from its effective APR spread. We refer to this as the “default-adjusted credit spread” and plot it on the right axis (black line) in Panel (b) of [Figure 2](#).

We find that, despite the relatively high charge-off rates, the default-adjusted credit spreads are positive and large. Thus, credit card borrowers pay rates that are a large spread over expected default losses. Indeed, the average default-adjusted credit spread (balance-weighted) is a very large 8.8% APR.

Panel (b) of [Figure 2](#) further shows that the default-adjusted credit spread is strongly decreasing in FICO score. It is close to 12% for FICOs below 660. There is a jump down to 10% between 660 and 670 FICO, likely related to this being the cutoff between non-prime and prime borrowers. The default-adjusted credit spread then decreases nearly linearly from around 10% at 670 FICO to 6% at 850 FICO.

Thus, although all borrowers pay a credit spread that is substantially larger than their expected credit loss, the gap is much wider for lower-FICO borrowers. This suggests that credit card rates may price in a large credit risk premium, and that this risk premium decreases in borrowers’ FICO score. We investigate this hypothesis in [section 5](#).

Next, we analyze accounts’ non-interest income and expense streams. We incorporate them into the calculation of returns on the FICO bins and ask to what extent they help explain the large default-adjusted credit spreads we find in Panel (b) of [Figure 2](#). We also analyze the returns banks earn on transactors, since their returns are generated exclusively by non-interest sources.

4.2 Non-interest Income and Expenses

The main non-interest components of credit card business’ profits are interchange income, rewards expense, fees, and operating expenses.

4.2.1 Interchange and Rewards

When a credit card is used to make a purchase, the merchant pays a “processing fee” or “swipe fee”, which is split between the bank that issued the credit card, the credit card network (Visa, Mastercard, American Express, or Discover), and the “payment processor”.

In the US in 2023, these three components of credit card processing fees amounted to \$162.5 billion (CMSPI, 2024). The bulk went to interchange fees, the component received by the card-issuing bank. In our data, we find that interchange fees average 1.82% of the purchase price. The network fee, which is paid to the credit card network, is typically lower—around 10-20 basis points of the purchase price for Visa and Mastercard (CMSPI, 2024).²²

A striking feature of credit cards is the prominence of rewards payments. Rewards are paid to the cardholder by their bank in return for using the card. Like interchange, they are also a percentage of the purchase price. Rewards are paid as cash, airline miles, or points that can be redeemed for cash or travel.

Rewards and interchange are closely intertwined. Rewards expenses are covered by banks' interchange income, which is why bank filings normally report the net of interchange income and rewards expenses as a single value. We find that banks' rewards costs are on average 1.57% of purchase volume. Hence, on average, banks pass on 86% of their interchange income as rewards to card users.

We compute the interchange and reward rates from the Y-14's portfolio-level data rather than the account-level data. While the account-level data contain a running sum of unspent rewards, it does not allow us to distinguish newly accrued rewards from those that are spent. To ensure that rewards are measured consistently and aggregate correctly, we use the portfolio-level data. The portfolio data report the dollar amounts of interchange income and rewards for each bank in each month. We calculate bank-level purchase volume by aggregating the purchase volume of the individual accounts during the month. We then compute the interchange and reward rates by dividing the dollar value of interchange (or rewards) for the bank by the bank's purchase volume. Appendix A.1 provides a detailed description of how we compute both rates.

Since interchange and rewards are proportional to purchase volume, we examine how purchase volumes vary across FICO scores. We separate borrowers and transactors since we are interested in measuring banks' return on assets for each group. Since transactors do not pay interest, the ROA for them depends much more on net interchange income.

²²The total interchange and network fee charged on Visa cards is between 1.51% and 2.5% of the transaction value, plus \$0.10 per transaction for transactions that take place in person. (<https://www.clearlypayments.com/interchange-rates-in-usa/>). For “card not present” (i.e., online) transactions, rates are higher, between 1.89% and 2.5% of the transaction value, plus \$0.10 per transaction. Mastercard's fees are similar.

Panel (d) of [Figure 1](#) plots the average purchase volume for borrowers and transactors in each origination FICO bin. We average only active accounts, i.e., those that have some purchase volume, payment activity, or balance that month. By construction this includes all borrowers, but it excludes inactive transactors.²³ There is little to no cost to the consumer of keeping an inactive account open, as long as it does not have an annual fee. We find that 22.6% of all accounts in our sample are inactive in a given month. Of the active accounts, on average 57.1% are borrowers and 42.9% are transactors.

Panel (d) of [Figure 1](#) reveals two main patterns. The first is that average purchase volumes are higher for active transactors than for borrowers across all FICO scores. This is not surprising since transactor accounts have no debt and hence are likely to have more capacity to spend.

The second pattern is that average purchase volume increases in FICO score for both transactors and borrowers. For transactors, purchase volume rises from a little over \$600 per month for 600 FICO to about \$1200 for 720 FICO, and \$1600 per month for FICO of 770 and above. For borrowers, purchase volume averages \$200 for 600 FICO, rising to \$600 for 720 FICO, and \$1300 by 850 FICO. This is not surprising either; FICO score and income are strongly positively related, hence accounts with higher FICO scores tend to have more purchasing power.

To understand how interchange and rewards contribute to the overall profitability of credit card portfolios, we compute lifetime interchange and reward as a fraction of ADB following the same procedure as above for interest income and charge-off losses. [Figure 3](#) plots the cross-section of these rates. As above, the rates are annualized. Unlike interest income and charge-offs, interchange fees and reward expenses apply to both transactors and borrowers. We plot them separately.

Panel (a) of [Figure 3](#) plots the results for transactors. Note that interchange and rewards are plotted on the left axis, while the difference between them, net interchange, is plotted on the right. The figure shows that for transactors, the interchange income (yellow) is close to 44% of ADB on an annualized basis for FICO scores between 600 and 800. Above 800 the average drops to around 33%. Rewards expenses (purple) are about 30-40% from 600 to

²³We exclude inactive accounts because our analysis focuses on the average purchase volume of actively used cards. This does not affect our other analysis, because inactive accounts do not take up space on the balance sheet. Inactive accounts may impose some administrative costs on the bank. These are included in operating costs, which we amortize over all active accounts.

800 FICO.

This means that the interchange income generated by transactors for the bank is akin to charging an annualized rate of 38% on the funds it (implicitly) provides them, though the payment is made by merchants, not the cardholders. This is a very high rate. To see where it comes from, note first that the average interchange fee is about 1.82% of purchase volume. The ratio of average purchase volume to average daily balance (ADB) for an average active transactor (FICO at 750) is about 1.8, so interchange revenue amounts to roughly 3.2% of ADB. Annualizing this figure (by multiplying by 12) gives the average annual rate of 38%. Similar calculations imply that the annual reward expense rate is 34.15% of ADB on average.

While a majority of interchange income is passed through to cardholders as rewards payments, net interchange income from transactors is substantial. Panel (a) of [Figure 3](#) shows that between 640 FICO and 810 FICO the average net interchange rate is about 4.6% of ADB annually. Thus, due to interchange, banks earn an annual rate of about 4.6% per dollar of financing they provide to transactors for their purchases.

For borrowers, interchange and rewards rates per dollar of ADB are much lower than for transactors. Panel (b) of [Figure 3](#) shows that average annual interchange income per dollar of ADB is flat at around 2.6% for FICO scores below 740, and then increases monotonically to 7.5% by 850 FICO. This increase is mainly due to a higher ratio of purchase volume to ADB for the high-FICO accounts. There is a similar pattern for rewards expenses: they are about 2% for FICO scores below 700, then rise to 7.4% by 850 FICO.

Net interchange income annually per dollar of ADB (right axis) is also much smaller for borrowers. It is uniformly less than or equal to 0.8% per year, and is between 0.35% and 0.6% for FICO between 620 and 815. For the highest FICO scores net interchange falls below 0.2%. The decrease is due to premium credit cards paying higher rewards rates. Such premium cards are mainly held by higher income, higher FICO-score cardholders.

Thus, [Figure 3](#) shows the net interchange income earned from transactors is large relative to assets. As we show below, it accounts for the bulk of the return on assets from transactors. In contrast, for borrowers it is a marginal component of return on assets compared to interest income and the default-adjusted credit spreads. Nevertheless, net interchange is positive for borrowers. Thus, rewards do not explain credit cards' high APRs; to the contrary, since net interchange is positive, it adds to the large default-adjusted credit spreads and hence deepens the puzzle of why credit card APRs are so high.

4.2.2 Fees

Credit card lending also generates other fee income besides interchange fees, including annual fees, late fees, balance transfer fees, over-limit fees, cash advance fees, convenience fees, and insufficient funds fees. During our sample period, such fees average 2.6% of credit card balances, accounting for approximately 10% of total credit card lending revenue.

The largest categories are late fees, annual fees, and balance transfer fees. Appendix [Figure C1a](#) plots average non-interest fees by FICO bin as a share of ADB during the life cycle of an account, starting from account origination. Fee income is seasonal, peaking every 12 months when annual fees are charged. Annual fees are highest for accounts in the highest and lowest origination FICO bins. At both extremes they are above 1% of ADB (not annualized).

Another significant source of fees is balance transfer fees, which accounts pay in order to transfer their borrowing from another credit card. These fees are highest for the lowest FICO accounts. Specifically, the 600 and 650 FICO accounts pay transfer fees of, respectively, 6% and 2% of ADB in the first month following origination.

[Table 2](#) reports income from fees as an annualized percentage of ADB across FICO bins for borrowers (panel A) and transactors (panel B). For borrowers, fee income is highest for subprime FICO scores (660 and below), driven by balance transfer fees and late penalties. Accounts with a 600 FICO score pay almost 5% annualized in fees on average, and even those with 640 FICO pay nearly 3.7%. For FICO scores between 680 and 800, fee income is flat at around 2%. For FICO scores above 800, fees rise again to nearly 2.9% for 850 FICO score, driven by the higher annual fees charged by premium cards, which are more common in this FICO range. The overall ADB-weighted average fee income for borrowers is 2.31% of ADB.

Fee income as a share of balances is high for transactors. Subprime accounts again generate the highest fees as a share of ADB, with annualized fee income of 6.6% to 14.1% of ADB. The high rate is due to annual fees and the relatively low ADB of these accounts (not to balance transfer or late fees, since by construction transactors do not incur these fees). For FICO scores of 680 and above, fee income is flat at around 4-5%. The overall ADB-weighted average fee income for transactors is 4.97%.

Adding interchange and fee income and subtracting reward expenses gives net non-interest income. Net non-interest income as a fraction of ADB is substantial for borrowers

and very large for transactors: for borrowers the ADB-weighted average is 2.7% of ADB, for transactors it is 9.6% of ADB. Since transactors do not pay interest, this is all of the income earned from them.

Adding non-interest income to the default-adjusted credit spread, we get the gross lending margin, the return lenders earn from borrowers after accounting for all income and variable expenses, but excluding operating expenses. We find that credit card lenders earn a very large 11.5% average gross margin on borrowers. Appendix [Figure C2a](#) plots the cross-section of gross margin by FICO bin. Like its components, it is decreasing in FICO score, from about 17.5% for a 620 FICO score to 9% for an 850 FICO. Thus, gross lending margin is large for all FICO scores—even the very safest borrowers generate a 9% gross lending margin. At the same time, gross lending margin varies substantially across FICO score, with higher-risk borrowers generating much larger gross margins.

4.3 Operating Expenses and the Significance of Marketing

Banks are not merely portfolios of financial assets and liabilities; most are also retail businesses that spend intensively on acquiring and serving retail customers. Credit card lenders exemplify this retail dimension of banking, as they engage extensively with consumers and incur large operating expenses on activities such as customer acquisition, applicant screening, card administration and customer service.

We illustrate this using the annual report (publicly reported) of a large credit card bank. The bank has three main business segments and provides disclosure of the operating expenses allocated to each, including its credit card operations. Not all banks break out these costs in this manner, making this institution’s detailed reporting useful for this illustration. We subsequently conduct a full sample analysis using Y-14 data.

This bank’s salaries and marketing are the largest operating expense components (Appendix [Figure C3a](#)). Among the bank’s overall operating expenses of \$20.3 billion, which include marketing expenses, \$12.5 billion is attributed to its credit card operations, representing 61.6% of the total (Appendix [Figure C3b](#)).²⁴ Since the credit card segment accounts for only 30% of the bank’s assets, it incurs a disproportionately large fraction of operating expenses.

²⁴The \$20.3 billion is also listed as “non-interest expense” in Appendix [Figure C3a](#). Adding up the non-interest expense for each of the three business segments confirms that they sum up to \$20.3 billion.

A comparison with the bank’s Commercial Banking segment highlights how expensive credit card operations are. The segment had an average of \$92.5 billion of loans (65% of the magnitude of credit card loans) and incurred \$2.0 billion in non-interest expense (16% of the corresponding figure for credit card loans). If Commercial Banking had incurred an equivalent cost per dollar of loans as credit card lending, its expenses would have been \$8.2 billion and it would have made substantial losses, given that its pre-tax income was only \$0.9 billion.

A good way to assess the magnitude of operating expenses is as a share of assets. For the Credit Card segment, operating expenses were 8.8% of assets. In other words, the bank had to make 8.8% per dollar of ADB just to recoup its operating costs. To put this into perspective, in 2023 operating expenses per dollar of assets for the aggregate US commercial bank system was 2.2%.²⁵ Thus, the credit card segment was four times as expensive to operate per dollar of asset as the average bank asset.

We find consistent results using the Y-14 data, which reports operating expenses specifically for the credit card segment reported by banks. The Y-14 operating expense includes spending on servicing, billing, card issuing, and authorizations. It does *not* include the costs of debt collections, fraud, and interchange processing, which are recorded separately. It does include spending on marketing, which is large, and which we discuss in detail below. We confirm that operating expenses are large: according to the Y-14 data they average 4.76% of balances for our sample, as shown in Panel C of [Table 2](#). Thus, accounting for operating costs offsets a substantial portion of card lending’s gross margin.

[Drechsler, Savov and Schnabl \(2021\)](#) show that banks pay significant fixed operating costs to gain market power in the retail deposit market. This enables them to borrow at rates that are both below the competitive short-term rate and are relatively insensitive to fluctuations in it. Since credit card banking is highly retail in nature, it is reasonable to expect that a similar mechanism is at work here. That is, credit card banks spend intensively on fixed operating expenses because it gives them substantial pricing power (a credit card franchise).

One way to examine this hypothesis is to analyze credit card banks’ spending on marketing. The role of marketing is to generate additional demand for a product (without changing

²⁵Even 2.2% is arguably a very high level of operating expenses for financial services. In comparison, the average expense ratio in 2024 for equity and bond mutual funds was 0.40% and 0.38%, respectively, money market funds charged 0.22% on average, and equity and bond index mutual funds charged only 0.05% on average (see [ICI Research Perspective, March 2025](#)).

the actual product itself). If the firm has pricing power, this additional demand generates a higher marginal profit by increasing the markup the firm can charge for a fixed supply.

In Appendix B we construct a model of the credit card market that captures this mechanism. There are two types of credit card borrowers in the model: those who are highly responsive to marketing, and those who are not. Banks pay a fixed operating cost to enter the market. Borrowers perceive cards as differentiated, so banks have some pricing power. In equilibrium, some banks optimally choose to also pay a fixed cost to engage in marketing, while the other banks do not. Marketing attracts the marketing-responsive borrowers. This increases marketing banks' market share, allowing them to charge a higher interest-rate markup, which they need to do to cover their greater per-dollar operating cost. Non-marketing banks optimally charge a lower markup and hence attract the borrowers who are unresponsive to marketing.

We find that marketing is a large component of operating expenses for credit card banks. Consequently, they are some of the biggest marketers in the US. For instance, in 2025, Capital One and American Express spent \$5.9 billion and \$6.3 billion on marketing, respectively; in 2023 their advertising spending made them the eighth and sixth largest advertisers in the US (see [Statista](#)). Thus, their marketing expenditures exceeded those of international consumer product giants Nike (\$4.7 billion) and Coca-Cola (\$5.4 billion). They also spent more than the largest banks: the second-largest US bank, Bank of America, spent \$2.2 billion in 2025, despite having 5.7 times the assets of Capital One, the largest US bank, JP Morgan, which had 7.5 times the assets of Capital One, spent \$5.5 billion.

Using the Call Reports, we compare the marketing expenses of “credit card banks,” which we define as those having a relatively high share of loans in credit card lending, with those of all other commercial banks. Appendix [Figure C4](#) plots the result. Credit card banks spend between 1-2% of their assets annually on marketing, compared to less than 0.1% for all other banks. The estimate for credit card banks is actually a lower bound, since even banks that meet our classification have a substantial share of non-credit card assets, including Capital One. If we were able to isolate the marketing expenses that are due specifically to the credit card business, their share of credit card assets would undoubtedly be higher. The fact that marketing's share of assets is more than ten times higher for the credit card business suggests that it is an effective means of generating demand, reflecting the highly retail, consumer-oriented nature of this business.

Next, we use the Y-14 data to analyze the relationship between the credit card banks' operating expenses per dollar and their gross margins. [Figure 4](#) plots the ADB-weighted relationship between banks' interest spreads (or gross margins) and their operating expenses per dollar share of ADB, within different origination FICO scores. The figure shows that controlling for the same ex-ante riskiness (i.e., within the same FICO bin), banks that have higher operating expenses per dollar of ADB charge higher interest spreads (Panel A) and earn greater gross margins (Panel B). This is particularly true for the banks that have large credit card operations.

We analyze this relationship formally by estimating the following weighted cross-sectional regressions:

$$Y_{b,f} = \alpha + \beta \text{Operating Expense}_b + \delta_f + \varepsilon_{b,f}$$

where the outcome variable, $Y_{b,f}$, represents lifetime borrower interest spread, net charge-off rate, or gross margin, aggregated across all observations at the bank (b)-origination FICO bin (f) level. Operating expense $_b$ is the bank-level average operating expense rate from the Y-14 portfolio-level data. We include origination FICO bin fixed effects, denoted by δ_f , to control for credit risk. Standard errors are clustered at the bank level.

Consistent with our hypothesis, column (1) in [Table 3](#) shows that, within an origination FICO bin, higher operating expenses are associated with higher interest spreads. In terms of economic magnitudes, a 1% increase in a bank's operating expense rate corresponds to a 62 basis point increase in the interest spread charged to borrowers. In column (2), we find no significant relationship between operating expenses and ex-post default rates when controlling for ex-ante credit scores, indicating that banks are not spending the higher operating expenses on better screen borrowers for default risk beyond their credit score.

Finally, column (3) shows that a 1% increase in the operating expense rate is associated with a 1.13% rise in the overall gross margin. This is consistent with the prediction of the model in [Appendix B](#) that banks' markups increase one-for-one with their per dollar operating expenses. Thus, an important factor in explaining credit cards' large interest spreads are the underlying banks' high operating expenses, including their substantial spending on marketing.

We have now covered all of the main components of credit card banks' income and expenses. In the next section, we combine these to calculate their return on assets for each FICO bin.

4.4 Putting it all Together: Return on Assets

Adding together all of an account’s streams of income and expenses per dollar of ADB over its lifetime gives us the banks’ lifetime return on assets (ROA) for the account. Thus, ROA equals interest spread minus net charge-offs (default-adjusted credit spread), plus net interchange income (interchange minus rewards), plus the fee income rate, minus the operating expense rate. The last column of [Table 2](#) reports the ROA across FICO bins and borrower/transactor groups.

[Figure 5](#) (right axis) plots the lifetime ROA for borrowers with different origination FICO scores. ROA is high and increases significantly with credit risk (decreases with origination FICO score), rising from 5% for borrowers with an origination FICO of 800 to 9% at 660 and 11% at 625. The magnitude of the increase in ROA with expected default probability is similar to the one we found for default-adjusted APR.

For transactors, ROA remains flat at 2.8% between FICO scores of 680 and 800 ([Appendix Figure C6](#)). The lower ROA observed at higher FICO scores is driven by higher rewards costs, while the higher ROA among riskier subprime accounts is due to higher fees and lower ADB relative to purchase volume, as they apparently transact closer to the end of the month and hence take up less balance sheet.

Overall, the ROA for transactors is mostly constant across FICO score and is on average slightly less than the ROA for even the safest borrowers. This is expected, since by construction transactors carry no credit risk. The contrast between the flat relationship of ROA and FICO score for transactors and the steep one for borrowers is another indication that credit card pricing incorporates a substantial risk premium component.

The high ROA of credit card lending is reflected in the aggregate ROA of banks specializing in this sector.²⁶ [Appendix Figure C12](#) presents a time series of ROA for credit card banks and other commercial banks based on Call Report data since 2001.²⁷ Call Reports allow us to extend the analysis further back in time to observe the performance of credit card lenders

²⁶According to the [FDIC Quarterly Banking Profile for 2024](#), the average pre-tax ROA of credit card banks in 2024 was 3.86% (ROE of 28.18%), compared to an average ROA for all FDIC-insured institutions of 1.4% (ROE of 11.37%).

²⁷We define credit card banks following the Federal Reserve’s Report to the Congress on the “Profitability of Credit Card Operations of Depository Institutions”, which classifies credit card banks as those where: (1) more than 50% of total assets are loans to individuals, and (2) at least 90% of consumer lending is related to credit cards or similar plans.

during the Global Financial Crisis. We compute ROA in the same way as account-level ROA, except we substitute the “provision for credit losses” in place of realized charge-offs to match the timing of net income (by construction the two series must cumulate to the same value over time).

Over the past 25 years, credit card banks have maintained an ROA of around 4%, roughly three times that of non-credit-card banks. To show how these bank-level estimates from the Call Reports compare to the (more precise) portfolio-level Y-14M-based estimates, Appendix [Figure C12](#) compares the two series for the post-2015 sample, when both are available. In the overlapping sample, the ROA calculated from the Call Reports co-moves strongly with the ROA calculated from the Y-14 portfolio-level data. However, the Call-Report-based bank-level ROA is lower than the ROA for the credit card portfolios, because credit card banks also operate other business segments, and these have lower ROAs.

The figure shows that during the GFC, credit card banks’ ROA declined sharply, falling to around zero in 2008 due to large credit losses. This is the only period in the sample during which credit card banks actually realized net losses, albeit small ones. Net income also turns negative during Covid-19 because banks recorded large loss provisions (provisions are expected losses; they reduce net income even though the losses have not yet been realized). However, these losses did not materialize (see e.g., [Zhou \(2025\)](#)) and the provisions were quickly reversed. Despite the high charge-offs during the GFC, credit card banks’ ROA was higher than that of other banks during this period. This is because the interest spread (risk premium) priced into credit card lending was large enough to absorb most of the increase in average charge-offs, even during the GFC.

5 Credit Card Risk Premium

As was shown in the previous section, the ROA of the credit card portfolios is closely linked to the default risk of their borrowers, as accounts with lower FICO scores (higher default risk) have substantially higher ROA. This suggests that the high return to credit card lending is due in part to a default risk premium. In particular, if credit card charge-offs tend to be higher during economic downturns or periods of heightened financial stress, then banks should demand compensation for bearing this risk.

The risk premium arises naturally in the formal model of credit card lending we present

in Appendix B. It arises due to the covariance between the default probability and the stochastic discount factor, as default is more likely in a downturn, which is when the SDF is high. As a result, the priced (i.e., risk-adjusted) probability of default and expected loss rate are higher than their objective/empirical counterparts, which creates a risk premium.

If there is a risk premium priced into credit card rates, then an important further question is whether it is consistent with the one priced into other assets that are exposed to heightened defaults in downturns.

Figure 6 illustrates the time series of charge-off rates across a variety of types of loans and corporate bonds. Panel (a) shows that credit card charge-offs strongly correlate with default rates on other loan types, especially during recessions. Notably, credit card charge-off rates are the highest and most sensitive to the cycles that occur in defaults. The commonality in defaults, and the fact that the common component spikes in recessions, suggests that this common default risk is undiversifiable (i.e., systematic) across loan markets. It is therefore reasonable to expect and may therefore be priced.

Panel (b) of Figure 6 focuses on the relationship between speculative-grade corporate bond defaults and credit card charge-offs. The series exhibit strong co-movement and comparable default rates.²⁸ This comparison suggests that corporate bonds can serve as a benchmark for assessing the risk premium in credit card lending. Corporate bonds are an especially useful benchmark because they are traded in secondary markets and are very large. Indeed, the corporate bond market is the main market for pricing credit risk.

We have seen that sorting on origination FICO score corresponds to large cross-sectional differences in average default rates across credit card loans. Figure 7 shows that it also corresponds to large differences in the volatility of charge-offs over time. Panel (a) plots the time series of charge-off rates over the life cycle for different origination FICO portfolios from the 2016 cohort. The average level of charge-offs is decreasing in origination FICO score, as one would expect. The charge-off levels are also quite stable over the life cycle, though the onset of the Covid period confounds the time series around the 4-year (48 month) mark. During the Covid period, charge-off rates declined, reflecting the disbursement of Covid checks, which, as shown in Zhou (2025), led to a large decline in charge-offs, especially among the low-FICO accounts that account for most charge-offs.

²⁸However, corporate bonds have a relatively high recovery rate, around 40%, compared to the 15% recovery rate on credit card charge-offs.

The other stand-out characteristic of the life cycle of charge-off rates is that they spike early in the life cycle for all FICO scores. The spikes correspond to the time when the promotional APR period expires, at which point borrowers are charged their regular APR rate.

Panel (b) of [Figure 7](#) shows the time series evolution of credit card charge-offs across origination FICO scores. Charge-offs are highly correlated across FICO portfolios, rising and falling in tandem with the aggregate charge-off rate. This shows that credit card charge-offs contain a common, undiversifiable component. In addition, the figure indicates that lower FICO score accounts experience a steeper increase in charge-off rates when the aggregate charge-off rate increases. Hence, the lower the FICO score, the greater is an account's sensitivity to aggregate charge-off risk.

To quantify the exposure of each FICO score portfolio to this aggregate charge-off risk, we follow the standard two-stage approach equivalent to [Fama and MacBeth \(1973\)](#). We first estimate a single-factor model of default risk using the cross-section of FICO portfolios. We proxy for this factor, the systematic component of charge-offs, with the quarterly change in the charge-off rate on the aggregate credit card portfolio. We then estimate the beta of each FICO portfolio to systematic charge-off risk by regressing the quarterly change in its charge-off rate on the factor:²⁹

$$\Delta \text{Charge-off Rate}_{i,t} = \alpha_i + \beta_i \Delta \text{Aggregate Charge-off Rate}_t + \epsilon_{i,t}. \quad (1)$$

[Figure 8](#) plots the estimated betas of the FICO portfolios against their corresponding charge-off rate. The estimates reveal a strong (one-) factor structure in the charge-off risks of the FICO portfolios, with charge-off beta strongly and linearly decreasing in FICO score. Thus, in addition to being a proxy for an accounts' expected default rate, FICO score is also a good proxy for its beta to systematic charge-off risk. For instance, low FICO portfolios (600-650) have high net charge-off rates, between 8-10% per year, and are strongly exposed to fluctuations in aggregate charge-offs, with estimated betas between 1.3 and 1.6. In contrast, high FICO portfolios (800-850) have much lower net charge-off rates, between 1-3%, and

²⁹In the estimation procedure, we exclude the first 18 months following account origination. Early charge-offs are not informative about systematic risk exposure because accounts generally cannot be charged off within 180 days of origination, and promotional terms early in the account's life (e.g., introductory APR offers) weaken the relationship between charge-off behavior and aggregate credit conditions.

charge-off risk, with estimated betas between 0.2 and 0.6.

Second, we use the betas to estimate the compensation for exposure to default risk, i.e., the default risk premium. To do so, we estimate a pooled panel regression of the ROA of FICO portfolio i in quarter t on its risk exposure, β_i , weighting observations by portfolio balances³⁰:

$$ROA_{i,t} = \lambda + \beta_i\gamma + \nu_{i,t}. \quad (2)$$

Thus, γ is the compensation for exposure to systematic default risk, i.e., the default risk premium, and λ is the average ROA on a credit card account that has a zero beta to systematic default risk.³¹

Figure 9 and Table 4 present the ROAs, estimated risk premiums $\beta_i\gamma$, and fitted ROAs ($\lambda + \beta_i\gamma$) for each FICO portfolio based on equation (2). The results show that the one-factor pricing model captures the average ROAs very well, with the model’s fitted ROAs aligning closely with actual average ROAs across the entire range of FICO scores. Hence, exposure to aggregate default risk appears to fully explain the strong, decreasing relation between FICO score and ROA.

We find that the price of default risk, γ , is a highly significant 4.3% per year. Consequently, the estimated risk premium $\beta_i\gamma$ ranges from 0.8-2.0% for the highest FICO portfolios (800-850) to 5-9.0% for the lowest FICO ones (600-650). Thus, the risk premium accounts for an increasing portion of the ROA as FICO score decreases.

The intercept λ represents the regression’s estimate of the ROA of a hypothetical zero-beta borrower, i.e., a borrower that has no exposure to systematic default risk. The estimate is 2.66%. Notably, this estimate is very close to the 2.57% ROA of the transactor portfolio, which carries no default-risk (since it does not involve borrowing) and thus has a zero beta by construction. Note that transactors were not included in the regression, and that they generate income through interchange and fees rather than interest charges. Thus, the similarity between their ROA and the estimated zero-beta rate supports the view that average ROA in excess of λ is compensation for charge-off risk.

Next, we examine how our estimates of credit card default risk premiums compare to those

³⁰Unweighted regression has minimal change in the risk premium estimates.

³¹We cluster standard errors by time and origination FICO score bin in the pooled panel regression. Our results are robust to using the Newey-West correction with an optimal number of lags to account for potential autocorrelation and heteroskedasticity in the residuals.

observed in other markets. We focus on corporate bonds because the market is large and economically important, and because default rates on corporate bonds are highly sensitive to the default cycle. Corporate bond credit spreads, defined as the difference between the yields on corporate debt and Treasury rates, have been found to be wider than implied by expected default losses alone. This phenomenon, known as the credit spread puzzle, has been attributed to a credit risk premium that compensates investors for increased credit risk during economic downturns (e.g., [Collin-Dufresne, Goldstein and Martin, 2001](#); [Amato and Remolona, 2003](#)).

To estimate the corporate bond credit risk premium, we use Mergent FISD historical data on corporate bond issuance from 1990 to 2023. For each bond, we track its history, including issuance details (date, amount and coupon), and whether and when the bond matured, defaulted or was called by the issuer, and use this to construct the cash flows paid to investors. In case of default, we assume a recovery rate of 40%, consistent with Standard & Poor’s (S&P) estimates.

Analogous to our approach for credit cards, we form monthly portfolios of bonds based on their date of issuance and initial credit rating. We then compute the rating portfolios’ buy-and-hold-to-maturity returns. To adjust the returns for the corresponding risk-free return, we subtract from each bond’s return the yield of a maturity-matched Treasury as of the bond’s issuance date. This gives us a monthly time series of default-adjusted credit spreads for each rating portfolio. The average of this time series is our estimate of the bond rating’s default-adjusted credit spread. Our sample covers dollar-denominated corporate bonds that were issued and matured during 1990-2023.

To facilitate the comparison between bonds and credit cards, we map corporate bond ratings to FICO scores based on the 5-year expected default rates provided by S&P and our own calculations for credit cards. [Figure 9](#) plots the credit card risk premiums estimated from the one-factor model in Equation (2) (black line) and the default-adjusted credit spreads for the corporate bond portfolios (green line). Remarkably, the corporate bonds’ default-adjusted credit spreads are very close to credit cards’ risk premium across most of the ratings range.

The exceptions are the lowest-rated bonds, those with a CCC/C rating. Their default frequency corresponds to a FICO score of around 620, at which credit cards’ risk premium exceeds the bonds’ default-adjusted credit spread by a substantial 3%.

This gap opens up because there is a concave relationship between bonds’ default frequency (which we map to a FICO score equivalent) and their estimated default-adjusted credit spreads. In contrast, the relationship between credit cards’ FICO scores and their estimated risk premium is effectively linear (due to the near-linear relationship between FICO score and default beta). Thus, bond credit spreads are slightly higher than credit cards’ risk premium for the highest rating (BBB), are the same at intermediate ratings (BB), and are lower at the lowest rating (CCC/C).

One caveat is that the credit card risk premiums are likely estimated much more precisely than the default-adjusted bond spreads. There are two reasons for this. First, we have a far finer gradient of FICO scores than bond ratings; there are 50 FICO bin portfolios, one for every 5 FICO point increment between 600 and 850, compared to only 4 bond rating portfolios. Second, there are many more credit card accounts than bond issuers. Our sample contains hundreds of millions of unique credit card borrowers, versus 10,025 unique bond issuers, of which only 540 were issuers of CCC/C-rated bonds. Thus, there is much more idiosyncratic noise in the bond portfolio returns, especially the lowest-rated ones where large individual defaults can affect the average.

Nevertheless, it is unlikely that noise accounts for the whole 3% gap between the default-adjusted spread of CCC/C-rated bonds and the corresponding credit card risk premium, as we are not aware of any estimate of the CCC/C default-adjusted spread that places it close to the estimated 8% risk premium on the corresponding credit cards. Thus, our results suggest that, compared to the corporate bond market, credit card banks likely earn an excess risk premium on the low FICO-score accounts.

Finally, we compare the zero-beta rate of credit cards, which represents their return adjusted for default risk, to the ROA of the aggregate U.S. banking sector. Using Call Report data over our sample period, we estimate the banking sector’s pre-tax ROA to be approximately 1.5%. Comparing this to the 2.66% zero-beta rate implies that credit cards yield a 1.16% higher default-risk-adjusted ROA than the ROA of the overall banking sector. This estimate represents a lower bound on this difference because the banking sector’s ROA has a small default risk exposure. Across commercial banks, average charge-offs amount to about 0.25% of total assets, or 0.5% of loans and leases. Based on the relationship between average charge-offs and default-risk betas estimated from the FICO portfolios in [Figure 8](#), we estimate that aggregate bank assets have a charge-off beta in the range of 0.05 to 0.1, which

maps to a default-risk premium of approximately 0.22% to 0.43%. Using this estimate implies that credit cards' zero-beta rate is 1.38% to 1.59% higher than the default-risk-adjusted ROA of the overall banking sector.

6 Extensions and Robustness

6.1 Risk Premia Using ROA Betas

Section 5 estimates the credit card default risk premium using the beta of each FICO portfolio's charge-off rate to the charge-off rate on the aggregate credit card portfolio. This approach focuses on systematic shocks' variation in defaults, which is the main source of shocks to the profitability of credit card portfolios. However, in principle, there may be other systematic shocks to credit cards' cash-flows, such as to fee income or net interchange. Moreover, if such shocks exist, exposure to them may also be priced.

To address this possibility, we estimate betas and risk premiums using portfolios' whole ROAs instead of their charge-offs, and we proxy for aggregate risk with the quarterly change in the ROA of the aggregate credit card portfolio. Thus, for each FICO bin i we estimate the following:

$$\Delta\text{ROA}_{i,t} = \alpha_i + \beta_i \Delta\text{Aggregate ROA}_t + \epsilon_{i,t}. \quad (3)$$

Appendix Figure C7 compares the estimates of the ROA betas with those of the charge-off betas. We find that the two sets of estimates are very similar and are statistically indistinguishable across all FICO bins. To understand why, Appendix Figure C8 plots the time series of ROA and (the negative of) charge-offs for the aggregate credit card portfolio and several FICO bins. The series are very similar to each other, which shows that fluctuations in ROA are mainly due to fluctuations in charge-offs, rather than the other components of ROA. As a result, portfolios' ROA and charge-off betas are very similar.

Finally, we re-estimate the credit card risk premiums using the ROA betas. The results are presented in Appendix Table D4. The estimated price of risk (4.64%), zero-beta rate (2.27%), and fitted ROAs (Appendix Figure C9) are very similar to, and statistically indistinguishable from, their corresponding charge-off beta counterparts. Thus, our estimates are effectively identical whether we estimate cards' risk using ROA or charge-off betas.

6.2 Representativeness of the Sample

A potential concern about our ROA estimates is that they may be biased upwards because the Y-14 sample (2015-2023) does not contain an episode in which there was a large spike in credit card charge-offs.³² In particular, the Y-14 sample excludes the Global Financial Crisis (GFC), the period during which credit cards experienced the most severe charge-offs. The question is whether including this period significantly changes our estimates of credit card ROAs, especially for the low FICO portfolios, since they are the most exposed to spikes in charge-offs.

To address this, we replicate the key elements of our charge-off analysis for the period 2008-2013, which covers the impact of the GFC, using data from the Office of the Comptroller of the Currency (OCC). Although the OCC dataset does not allow us to compute lifetime ROAs for accounts originated prior to the GFC and affected by the rise in charge-offs during the crisis, because the data begin only in 2008, it does allow us to analyze the main driver of risk premia—charge-off rates—during the most severe downturn in the history of the credit card market.

Appendix Figure C10 plots charge-off rates during the GFC across origination FICO portfolios. Consistent with Figure 6, default rates on credit cards surged to historically high levels during this time. Charge-offs exceeded 25% for accounts with an origination FICO score of 600, and surpassed 10% even for super-prime accounts, with FICO scores of 750. Consistent with the pattern in Figure 7 for the 2015–2023 sample, the low-FICO portfolios exhibit the greatest sensitivity (beta) of their charge-offs to aggregate conditions.

To estimate the impact of this subsample on portfolios’ average ROAs, we utilize the strong factor structure in credit card portfolios’ realized charge-offs and ROAs. Since charge-off and ROA betas are virtually identical (see section 6.1 and Appendix Figure C7), we use the FICO portfolios’ charge-off betas over 2008-2013, which we can estimate using the OCC sample, to proxy for their ROA betas in this sample. The shocks to portfolios’ average ROAs can then be estimated by scaling the shock to aggregate credit card ROA by their betas.

Appendix Figure C11 compares the charge-off betas of the FICO portfolios for the 2008–2013 OCC sample with the betas from our baseline Y-14 sample (2015–2023). The

³²The sample contains the Covid period, which was expected to experience severe charge-offs. However, as documented in Zhou (2025), this did not occur due to the government’s large stimulus payments, which many households used to pay down their balances.

estimated betas are remarkably similar across the two samples. This shows that FICO portfolios’ exposures to aggregate charge-off shocks are very stable, including when charge-offs spike. This suggests that betas should continue to be strongly related to FICO portfolios’ risk premiums across other samples.³³

Since the FICO portfolios’ betas are stable across samples, the remaining question is how much the estimated compensation for charge-off beta falls when we include the GFC subsample. Although we cannot directly measure the lifetime ROAs of accounts originated prior to the GFC and exposed to the rise in charge-offs in the OCC data, because the sample begins only in 2008, we approximate the impact of GFC losses on average credit card ROA using the average ROA of credit card banks from the Call Reports (see Section 2.2).

Appendix Figure C12 shows that the aggregate credit card bank ROA is highly correlated with the ROA of the aggregate credit card portfolio from the Y-14M data for the overlapping period 2015-2023. Although there is a difference in the averages of the two series—bank-level ROAs have a lower average because the credit card banks also hold lower-yielding non-credit-card assets—changes in the two series track closely.

We also find that credit card ROA did not decline substantially during the GFC, despite very high charge-offs. Appendix Figure C12 shows that the aggregate ROA of credit card banks fell sharply, becoming negative in 2008Q4 and 2009Q1 and staying low through 2010Q1. Yet, even then, average ROA barely turned negative despite the high charge-offs. This is because of the large default risk premiums priced into credit card lending rates, which implies that the credit card portfolio can absorb large unexpected charge-offs before its ROA becomes negative. As a result, credit card banks’ ROA did not fall substantially below the ROA of the rest of the commercial bank sector, even though credit card charge-offs have the highest sensitivity to downturns (however, the degree of under-performance of assets such as mortgages was likely more unexpected than that of credit cards during the GFC).

We use the credit card bank ROA series to assess the impact of including the GFC on estimates of average credit card ROA. During the Y-14 subsample (2015-2023), the average

³³Before the CARD Act, lenders had the flexibility to adjust the interest spread on existing borrowing. This flexibility could affect the beta estimates if changes in interest rates influence charge-off behavior differently across FICO scores during the crisis. However, we find no evidence of this given that the estimated betas from the pre-CARD Act sample are so similar to those from the 2015-2023 sample. This likely reflects the finding in Nelson (2025) that increases in the interest rate on existing borrowing have minimal effect on default rates.

ROA of credit card banks was 3.97%. If we extend the sample to 2008-2023, so that it includes the GFC, average ROA falls to 3.85%. This is because the average ROA over 2008-2014 is 3.70%. Thus, average ROA does fall when we include the GFC subsample, but the difference is very small. There are two reasons for this. The first is that, as discussed above, the fall in ROA during the GFC is not as dramatic as one might expect; ROA remains mostly positive even during the GFC, because of the large risk premium in credit card rates. The second is that in the immediate post-GFC years average ROA is relatively high, as shown in Appendix [Figure C12](#). Thus, banks earn back some of the returns lost during the GFC. The result is that including the GFC subsample has little impact on our estimate of average credit card bank ROA.

In summary, including the subsample that encompasses the GFC has little impact on our estimates of credit card portfolios' charge-off betas, or the large average compensation earned for this beta. We thus conclude that our estimates of the cross-section of risk premiums earned by the credit card portfolio are robust to including the GFC subsample.³⁴

6.3 The Impact of Capital Requirements

In this section, we investigate whether bank capital regulation can account for credit card portfolios' high ROAs. Under Basel III, banks must have minimum equity capital equal to a prescribed fraction of risk-weighted assets (RWA), in order to absorb potential credit losses. The regulation assigns risk-weights based on the risk of an asset, so banks must maintain more capital for riskier exposures. Cash and government bonds are safe, so they carry a 0% risk weight. In contrast, corporate loans and credit card debt are risky and hence have much higher risk weights.

If regulatory capital is costly for banks, higher risk weights on credit card loans could explain part of the spread between their ROAs and those of the average bank asset. Moreover, regulatory capital costs could also account for part of the decreasing relation between FICO score and ROA, since regulatory risk weights are decreasing in FICO score. On the other hand, capital regulation may play only a limited role in explaining the level and slope of

³⁴Credit card banks' average ROA was even higher in the 2001-2007 sample, at 5.73%. In this respect, this situation is similar to the rest of the commercial banking sector, which earned a relatively high average ROA of 1.95% over 2001-2007, compared to 1.39% during the Y14 sample and 1.16% over the 2008-2023 sample.

ROAs if credit card risk weights are not much larger than those of other bank assets, or if the cost of regulatory capital is small relative to the risk premium investors demand for bearing the risk of credit card loans. Thus, the importance of capital requirements in explaining these ROAs is ultimately an empirical question.

To address this question, we compute regulatory risk weights for credit card loans across the FICO distribution. Risk weights differ depending on whether a bank uses the standardized approach (SA) or the internal ratings-based (IRB) approach, also called the advanced approach. Large banks, with total consolidated assets above \$250 billion, typically use the IRB approach. This relies on internal estimates of probability of default (PD) and loss given default (LGD). Smaller banks typically use the SA approach.

Under the standardized approach, the risk weight for credit card loans is fixed at 100% for both borrowers and transactors (Code of Federal Regulations, 12 C.F.R. § 217.32). Under the IRB approach, risk weights depend on the bank’s internal PD and LGD estimates, following the Basel III regulatory formulas (Code of Federal Regulations, 12 C.F.R. § 217.131). The Y-14M records the account-level PD and LGD data. Using it, we compute risk weights separately for borrowers, transactors, as well as all accounts, by origination FICO bin.

Panel (a) of Appendix [Figure C13](#) plots the average risk weights assigned by SA and IRB banks as a function of FICO score. SA banks simply assign a constant 100% risk weight across all FICO scores, while IRB banks’ risk weights are strongly declining in FICO score. For accounts that are currently borrowers, the average risk weight assigned by IRB banks decreases from about 150% for a 600 FICO score to 35% for an 850 FICO score. For the IRB banks there is also a large difference between the risk weights assigned to accounts that are currently borrowers versus transactors. The average risk weight assigned to transactors is always less than half, and often less than one-third, of that assigned to borrowers at the same FICO score. Thus, the average transactor risk weight declines from about 60% for 600 FICO to 10% for 850 FICO.³⁵ Panel (b) plots the balance-weighted average of the risk weights across all banks.

Next, we conduct three exercises to analyze whether capital requirements can explain the level of ROA and its relationship with FICO score.

³⁵Transactors’ risk weights depend on FICO because they may become borrowers in the future. The fact that they are so much lower than borrowers’ indicates that they tend to remain transactors.

6.3.1 Do banks with larger risk weights earn higher ROAs?

Our first test exploits cross-sectional differences in risk weights between IRB and SA banks.³⁶ As shown in Panel (a) of [Figure C13](#), IRB banks face higher risk weights for borrowers below roughly 725 FICO, while SA banks face higher risk weights for borrowers above this threshold. The difference in risk weights rises to 50 percentage points for low-FICO borrowers and 70 percentage points for high-FICO ones.

If capital requirements materially shape pricing, banks with higher risk weights for a given FICO score should charge higher interest spreads and earn higher ROAs on those scores to compensate for the higher regulatory capital cost. We test this by estimating weighted cross-sectional regressions of the form:

$$Y_{b,f} = \alpha_f + \beta \text{Risk Weight}_{b,f} + \varepsilon_{b,f}$$

where $Y_{b,f}$ represents either average interest spread, net charge-off rate, or ROA earned by bank b on borrowers in FICO bin f . $\text{Risk Weight}_{b,f}$ denotes the average regulatory risk weight that bank b applies to borrowers in FICO bin f . In an alternative specification, we replace the bank-level value with the group-level average risk weight, i.e., the average risk weight assigned by all banks that use the same regulatory approach (IRB or SA) as bank b . In all specifications, we include origination-FICO fixed effects, which absorb all cross-sectional variation in borrower credit risk. Identification therefore comes from differences across banks (or bank groups) in the risk weights applied to borrowers with the same ex-ante credit risk, allowing us to isolate whether higher regulatory capital requirements lead to higher interest spreads and ROAs.

The results are presented in [Appendix Table D5](#). They show no evidence that banks facing higher risk weights charge higher interest spreads (columns (1) and (4)). We also do not find any difference in charge-off rates (columns (2) and (5)). Likewise, banks facing higher risk weights do not earn higher ROAs on their borrowers (columns (3) and (6)). These findings suggest that capital requirements play little role in shaping the cross-section of credit card rates or profitability.

³⁶This approach follows earlier work studying the impact of capital requirements on mortgage lending in the U.K. ([Benetton, 2021](#)) and corporate lending in Germany ([Behn, Haselmann and Wachtel, 2016](#)).

6.3.2 Comparing credit card ROA to the cost of regulatory capital

Our second exercise compares the ROA of credit card lending at each FICO score to an upper bound on the cost of required regulatory capital for it. We calculate the upper bound by assuming the entire ROA of the aggregate banking sector is compensation for the cost of regulatory capital. Dividing aggregate ROA by the banking sector’s risk-weighted assets—which determines the required regulatory capital—is then an upper bound on the cost of regulatory capital per unit of risk-weighted assets. Multiplying this upper bound by the risk weight for credit card lending at each FICO score then gives us an upper bound for its regulatory capital cost.

This is an upper bound because some of banks’ ROA is likely to be compensation for banks’ exposure to priced economic risks, rather than compensation for the cost of regulatory capital. For instance, the banking sector is exposed to systematic credit risk, as was clear during the GFC. More broadly, bank stocks have a market beta of around 1. Deleveraging this equity beta by banks’ (book) leverage of around ten ([FDIC Quarterly Banking Profile](#)) suggests an average asset beta of around 0.1. Standard compensation for market risk can then account for a large fraction of banks’ ROA.³⁷

Using the Call Reports, we compute the average risk weight of banks by dividing the banking sector’s total Risk-Weighted Assets (RWA) by total assets. We find that the average risk weight is 70%. This is higher than the risk weight for transactors and substantially lower than the risk weights for borrowers with low FICO scores.

From the Call Reports, we estimate banks’ pre-tax ROA to be 1.5%, which is confirmed by the figures reported in the [FDIC’s Quarterly Banking Profile](#). Dividing this value by the average risk weight of 70% gives a pre-tax ROA per unit risk weight of 2.14%. Multiplying this value by credit cards’ risk weight (Panel (b) of [Figure C13](#)) gives the upper bound on the cost of regulatory capital at each FICO score.

The values of the bound are plotted in [Figure C14](#), together with the ROAs for borrowers. The figure makes clear that credit cards’ ROAs are much larger, and decrease much more with FICO score, than the cost of their regulatory capital. Thus, the cost of regulatory capital

³⁷As of January 2026, ETFs tracking the banking sector have an estimated beta of 1.38 (KBE) and 1.49 (KBWB), while JP Morgan Chase, the largest bank, has a beta of 1.07. Assuming leverage of ten suggests that banks’ asset beta is roughly 0.1. A standard estimate of the market risk premium of 8% then implies that the average bank asset should earn a 0.8% market risk premium. This would account for around 80% of the banking sector’s *after-tax* ROA, which is approximately 1.0% ([FDIC Quarterly Banking Profile](#)).

does *not* explain much of either the level of ROA on credit card lending or its relation with FICO score. This result is not surprising once we recognize that credit cards’ risk weights are ‘only’ 40% to 130%, or 0.55 to 1.85 times the risk weight of the average bank asset, whereas credit card ROAs are 3-7 times the ROA of the average bank asset. Thus, credit card ROAs are much larger than required to meet the cost of regulatory capital, even under our loose upper bound.

6.3.3 Incorporating capital requirements into the risk premium estimation

Finally, we incorporate the regulatory capital requirement directly into our estimation of risk premia. We augment the second-stage regression by including each portfolio’s risk weight:

$$ROA_{i,t} = \lambda + \beta_i \gamma + \text{Risk Weight}_{i,t} \zeta + \nu_{i,t}. \quad (4)$$

Here, γ is compensation for exposure to systematic default risk (as in our baseline specification), while ζ captures the compensation for required regulatory capital.³⁸

Figure C15 and Appendix Table D4 report the results. Consistent with our previous findings, the estimated compensation for capital requirements is small. The estimate of ζ is 0.92%, and is not statistically significant. This value implies that the compensation for regulatory capital increases from roughly 0.37% for high-FICO portfolios to about 1.2% for low-FICO portfolios, or around 40% of the upper bound discussed above.

Including risk weight in the analysis slightly reduces the estimated risk premium for default exposure from 4.3% in the baseline to 3.9%, and lowers the estimated zero-beta rate from 2.66% to 2.17%. The fitted ROA curve remains virtually identical to the baseline case.

Taken together, all three exercises point to the same conclusion: capital requirements play at most a limited role in explaining the high ROA of credit card lending, since the cost of regulatory capital is too small to account for ROAs that are several times larger than the banking sector’s average.

³⁸Since regulatory risk weights depend on the expected default probability for a one-year horizon, they are correlated with the FICO portfolios’ exposure to systematic default risk factor (i.e., their charge-off beta). However, as shown in Panel (b) of Figure C13, the relationship between FICO score and risk weights is substantially flatter than the relationships between FICO score and charge-off beta or FICO score and ROA. The purpose of this specification is to assess whether allowing for explicit compensation for regulatory capital materially alters the estimated default risk premium or the fitted ROA profile. As we show below, including risk weights has a minor effect on the estimates.

7 Conclusion

Credit card interest rates are significantly higher than those on other forms of lending, with an average APR of 22% in 2025—an 18% spread over the federal funds rate. In this paper, we investigate why these rates are so high and, more broadly, how credit card banking operates. This question is particularly important given that credit cards are the primary method of unsecured borrowing for consumers, with 60% of accounts carrying a balance. With more than \$1.1 trillion in outstanding balances and generating 16.3% of banks’ interest income despite comprising only 4.5% of bank assets, understanding the factors driving credit card pricing has broad implications for financial markets.

To answer this research question, we use the most comprehensive and granular dataset, covering 550 million credit card accounts from the 20 largest banks in the US, representing over 90% of total credit card lending. We examine four potential explanations for high credit card rates: compensation for default losses, the cost of rewards programs, a large default risk premium, and market power. Our findings show that while charge-offs are substantial, they do not fully explain the high APR spreads; banks maintain an average default-adjusted spread of 8.8% after accounting for the defaults. Non-interest expenses, such as rewards, are substantial but offset by corresponding non-interest income, such as interchange. Instead, market power plays a significant role, with banks incurring large operating expenses—especially in marketing—to acquire and retain customers.

In addition, undiversifiable default risk is a major driver of pricing, as credit card charge-offs are highly cyclical and correlated with broader economic downturns. We estimate the risk-premium component of credit card return using the cross-section of accounts by default likelihood. We find that the risk premium can explain the large cross-sectional differences in net lending margins, with the average borrower commanding a 4.3% risk premium, similar to that in high-yield bonds. After adjusting for the risk premium, we find that credit card lending still generates “alpha” of approximately 1.5% compared to the overall banking sector.

Our findings highlight the distinct economics of credit card lending, where high interest rates reflect more than just default losses—risk premium and market power play a central role. Looking beyond this paper, several promising directions for future research remain. For instance, it would be fruitful to explore how competition and regulation influence credit card banking and how it interacts with emerging forms of consumer credit.

References

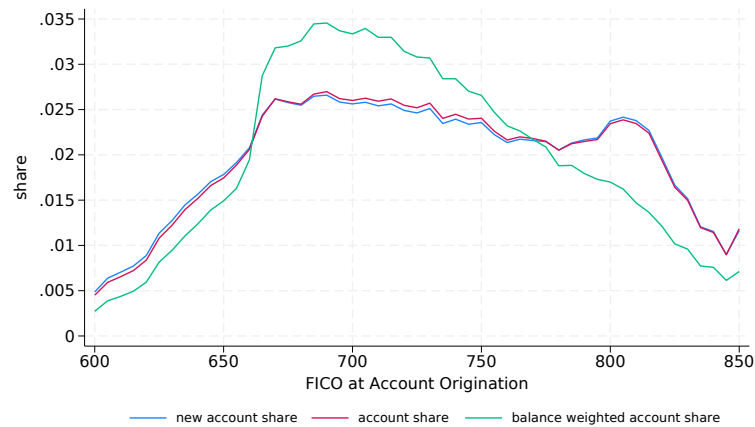
- Adams, Robert, Vitaly M. Bord, and Bradley Katcher. 2022. “Credit Card Profitability.” *FEDS Note*.
- Agarwal, Sumit, Andrea Presbitero, André Silva, and Carlo Wix. 2022. “Who Pays For Your Rewards? Cross-Subsidization in the Credit Card Market.” *Working Paper*.
- Agarwal, Sumit, Souphala Chomsisengphet, Neale Mahoney, and Johannes Stroebel. 2015. “Regulating Consumer Financial Products: Evidence from Credit Cards.” *The Quarterly Journal of Economics*, 130(1): 111–164.
- Amato, Jeffery D, and Eli M Remolona. 2003. “The credit spread puzzle.” *BIS Quarterly Review*, December.
- Ausubel, Lawrence. 1991. “The Failure of Competition in the Credit Card Market.” *The American Economic Review*, 81(1): 50–81.
- Behn, Markus, Rainer Haselmann, and Paul Wachtel. 2016. “Procyclical capital regulation and lending.” *The Journal of Finance*, 71(2): 919–956.
- Benetton, Matteo. 2021. “Leverage regulation and market structure: A structural model of the UK mortgage market.” *The Journal of Finance*, 76(6): 2997–3053.
- Berlin, Mitchell, and Loretta J Mester. 2004. “Credit card rates and consumer search.” *Review of Financial Economics*, 13(1-2): 179–198.
- Calem, Paul, and Loretta Mester. 1995. “Consumer Behavior and the Stickiness of Credit-Card Interest Rates.” *The American Economic Review*, 85(5): 1327–1336.
- CMSPI. 2024. “State of the Industry Report.” *White Paper*.
- Collin-Dufresn, Pierre, Robert S Goldstein, and J Spencer Martin. 2001. “The determinants of credit spread changes.” *The Journal of Finance*, 56(6): 2177–2207.
- Consumer Financial Protection Bureau. 2023. “The Consumer Credit Card Market.” *Report*.
- Cordell, Larry, Michael R Roberts, and Michael Schwert. 2023. “CLO performance.” *The Journal of Finance*, 78(3): 1235–1278.
- Dempsey, Kyle, and Felicia Ionescu. 2025. “Default Risk and Pricing in the U.S. Credit Card Market.” *Working Paper*.
- Drechsler, Itamar, Alexi Savov, and Philipp Schnabl. 2017. “The Deposits Channel of Monetary Policy.” *The Quarterly Journal of Economics*, 132(4): 1819–1876.
- Drechsler, Itamar, Alexi Savov, and Philipp Schnabl. 2021. “Banking on Deposits: Maturity Transformation without Interest Rate Risk.” *The Journal of Finance*, 76(3): 1091–1143.
- Drozd, Lukasz, and Jaromir Nosal. 2008. “Competing for Customers: A Search Model of the Market for Unsecured Credit.” *Working Paper*.
- Erel, Isil, Thomas Flanagan, and Michael S Weisbach. 2024. “Risk-Adjusting the Returns to Private Debt Funds.” National Bureau of Economic Research.
- Fama, Eugene F, and James D MacBeth. 1973. “Risk, return, and equilibrium: Em-

- pirical tests.” *Journal of political economy*, 81(3): 607–636.
- Flanagan, Thomas.** forthcoming. “The value of bank lending.” *Journal of Finance*.
- Fleckenstein, Matthias, and Francis A. Longstaff.** 2022. “The Market Risk Premium for Unsecured Consumer Credit Risk.” *The Review of Financial Studies*, 35(10): 4756–4801.
- Galenianos, Manolis, and Alessandro Gavazza.** 2018. “Regulatory intervention in consumer search markets: The case of credit cards.” Working paper.
- Gupta, Arpit, and Stijn Van Nieuwerburgh.** 2021. “Valuing private equity investments strip by strip.” *The Journal of Finance*, 76(6): 3255–3307.
- Guttman-Kenney, Benedict, and Andres Shahidinejad.** 2025. “Unraveling Information Sharing in Consumer Credit Markets.” *Working Paper*.
- Han, Song, Ben Keys, and Geng Li.** 2018. “Unsecured Credit Supply, Credit Cycles, and Regulation.” *The Review of Financial Studies*, 31(3): 1184–1217.
- Herkenhoff, Kyle, and Gajendran Raveendranathan.** 2021. “Who Bears the Welfare Costs of Monopoly? The Case of the Credit Card Industry.” *NBER Working Paper*, <http://www.nber.org/papers/w26604>.
- Kaplan, Steven N, and Antoinette Schoar.** 2005. “Private equity performance: Returns, persistence, and capital flows.” *The journal of finance*, 60(4): 1791–1823.
- Korteweg, Arthur, and Stefan Nagel.** 2016. “Risk-adjusting the returns to venture capital.” *The Journal of Finance*, 71(3): 1437–1470.
- Nelson, Scott.** 2025. “Private information and price regulation in the us credit card market.” *Econometrica*.
- Neumark, David, and Steven A Sharpe.** 1992. “Market structure and the nature of price rigidity: evidence from the market for consumer deposits.” *The Quarterly Journal of Economics*, 107(2): 657–680.
- Stango, Victor, and Jonathan Zinman.** 2016. “Borrowing High versus Borrowing Higher: Price Dispersion and Shopping Behavior in the U.S. Credit Card Market.” *The Review of Financial Studies*, 29(4): 979–1006.
- Zhou, Guanyu.** 2025. “The Worst and Best of Times: Stimulus and Credit Card Debt.” *Working Paper*.

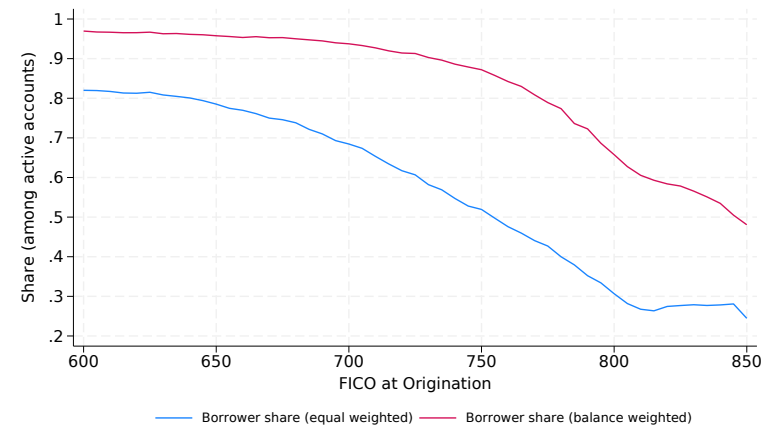
Figure 1: Accounts, balances, and purchase volumes by origination FICO score

This figure presents account characteristics across origination FICO bins constructed using 5-point intervals. Panel (a) plots the distribution of accounts. New account share (blue line) is the proportion of newly originated accounts out of the total number of accounts. Account share (red line) is the proportion of account-month observations in each FICO bin, and balance-weighted account share (green line) is the proportion of lifetime balances held by accounts in each bin. Panel (b) plots the borrower share based on the number of accounts (blue line) and total borrower balances (red line). Panel (c) plots Average Daily Balance (ADB, in dollars) for borrowers (blue line) and active transactors (red line). Panel (d) plots average monthly purchase volume (in dollars) for borrowers and active transactors. Active transactors are accounts with a positive cycle-ending balance, purchase volume, or payment activity in a given month.

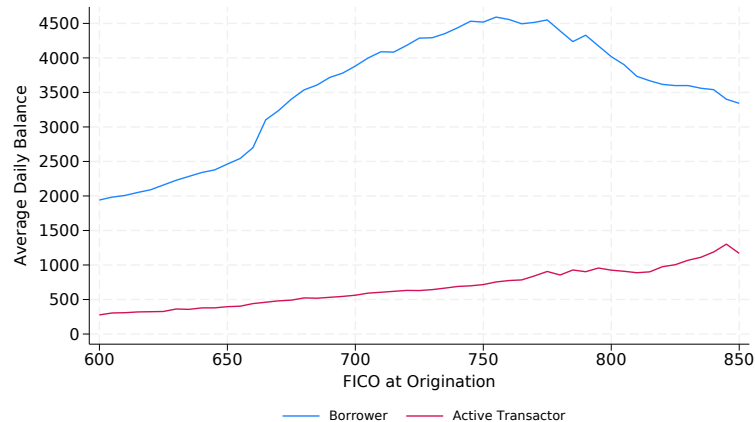
(a) Distribution of accounts



(b) Borrower share



(c) Average Daily Balance (ADB)



(d) Monthly purchase volume

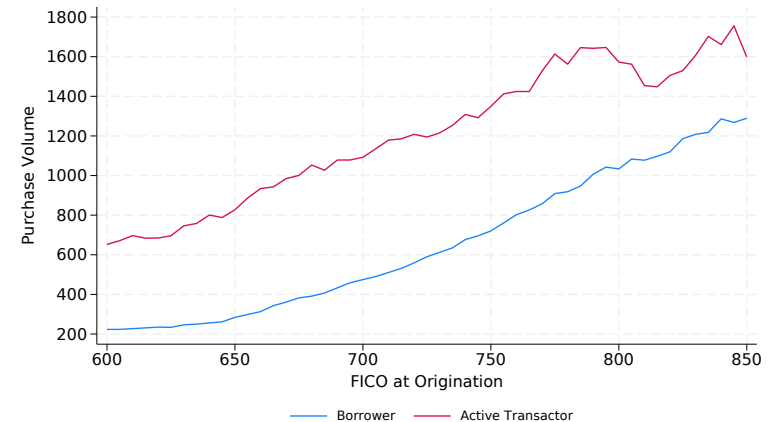
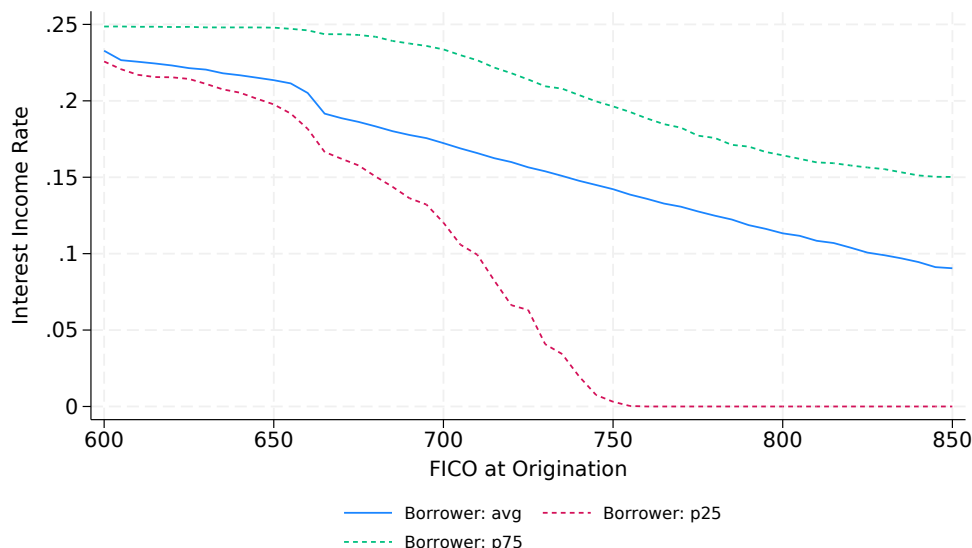


Figure 2: Effective interest rates and default-adjusted credit spreads for borrowing accounts

This figure presents effective interest rates and default-adjusted credit spreads for borrowing accounts by origination FICO score. Panel (a) shows the 25th percentile (dashed red line), average (solid blue line), and 75th percentile (dashed green line) of the effective interest rates paid by borrowers within each origination FICO bin. The effective interest rate is calculated as the reported finance charge divided by the borrower's Average Daily Balance (ADB), annualized by multiplying by 12. The average effective interest rate weights borrowers within each FICO bin by their ADB. Panel (b) plots interest income (blue area), interest expense (red area), and net credit losses (green area) as shares of ADB on the left y-axis, and the default-adjusted credit spread (black line) on the right y-axis. Within each FICO bin, we calculate lifetime ADB, interest income, interest expense, and net credit losses by summing each variable across borrowing accounts and over the sample period. Interest income is the account's finance charge, interest expense is the account's ADB multiplied by the monthly federal funds rate, and net credit losses equal charge-offs minus recoveries. The default-adjusted credit spread equals the interest income rate minus the interest expense rate and the net credit loss rate. All rates are annualized.

(a) Distribution of effective interest rates paid by borrowers



(b) Net interest income, charge-offs & default-adjusted credit spread

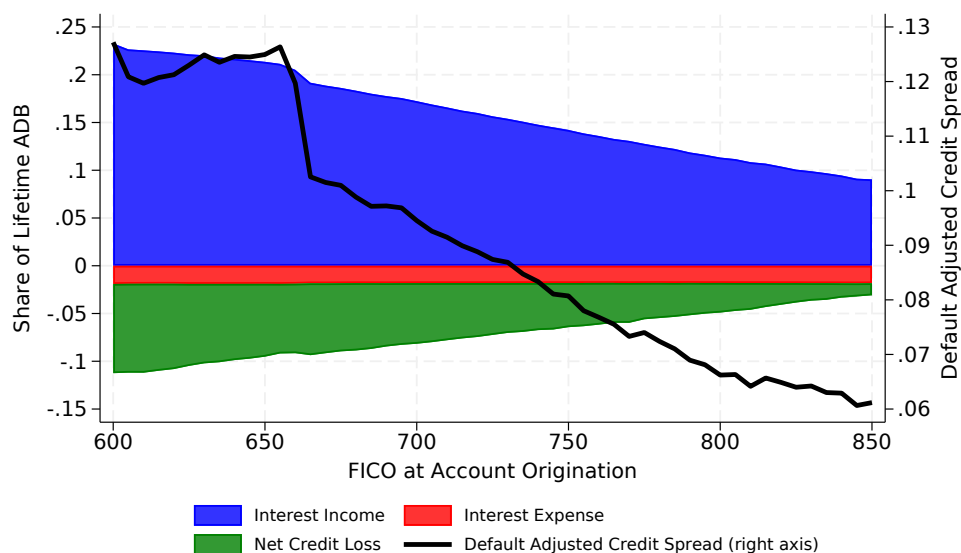
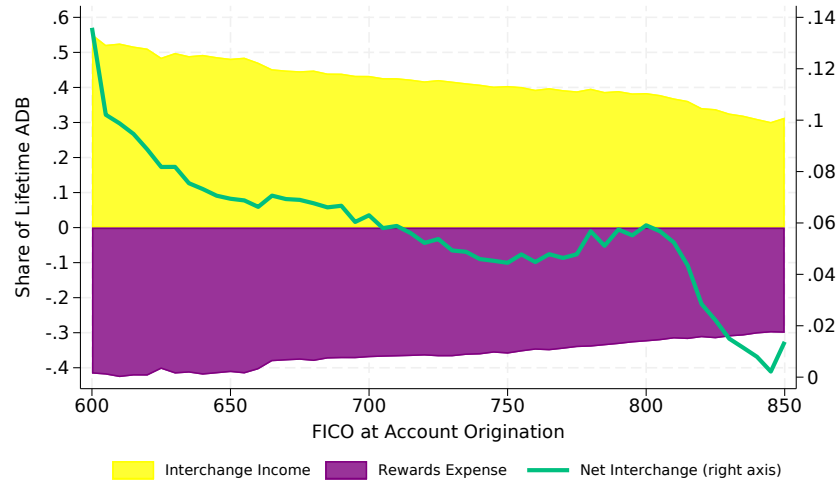


Figure 3: Interchange and rewards for borrowers and transactors

This figure plots interchange income (yellow area), and rewards expenses (purple area) as a share of ADB on the left y-axis, and net interchange income, defined as interchange minus rewards (black line), as a share of ADB, on the right axis. The variables are computed as follows. We first use the Y-14's account-level data to calculate bank-level purchase volume by aggregating the purchase volume of all individual accounts during the month. Second, we use bank-level information on total interchange income and reward expenses from the Y-14's portfolio-level data, and divide these amounts by the bank's purchase volume, calculated in the first step, to compute the interchange and reward rates. Third, we compute interchange income and rewards expense for each origination FICO score bin by multiplying the interchange and rewards rates derived in step two by the lifetime purchase volume of the accounts in that bin. Fourth, we divide the interchange income and reward expenses by the bin's lifetime ADB to compute their share of lifetime ADB. All values are annualized. Transactors are plotted in panel (a), borrowers in panel (b). Appendix A.1 provides a detailed description of the calculations.

(a) Transactors



(b) Borrowers

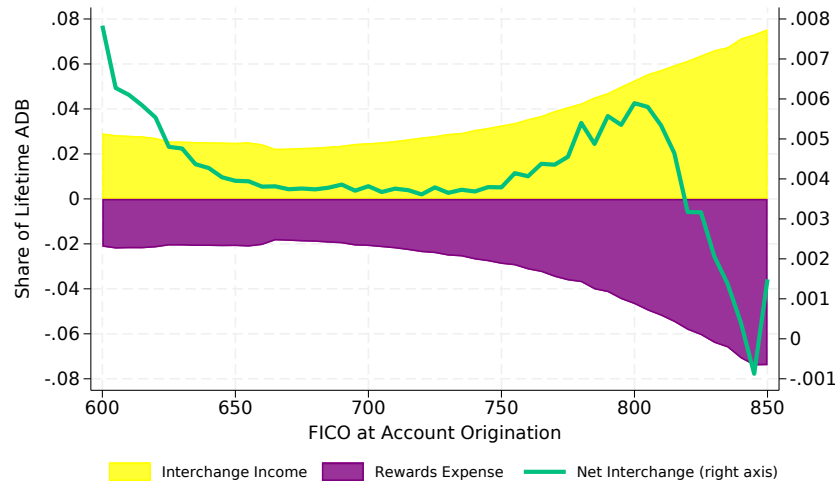


Figure 4: Relationship of bank operating expenses with borrower interest spreads and gross margins

Panel (a) presents a binned scatter plot of borrowers' interest spreads against bank-level operating expense rates. Borrower interest spread is calculated as total finance charges minus interest expenses across all borrower observations within a bank-origination FICO bin, divided by the total of borrowers' Average Daily Balance (ADB) for that bin. The operating expense rate is bank's credit card operating expenses divided by its total cycle-ending balance, measured at the bank-month level and averaged over the sample period from January 2015 to December 2023. The red line is the fitted line from regressing borrowers' interest spread rate on their bank's operating expense rate, controlling for borrower's origination FICO score. The regression is weighted by borrowers' ADB. Panel (b) follows the same structure as Panel (a) but replaces interest spread with gross margin as the outcome variable. Gross margin is the net interest spread plus net interchange and fee income, minus net charge-offs and other non-operating expenses (e.g., fraud). All rates are annualized.

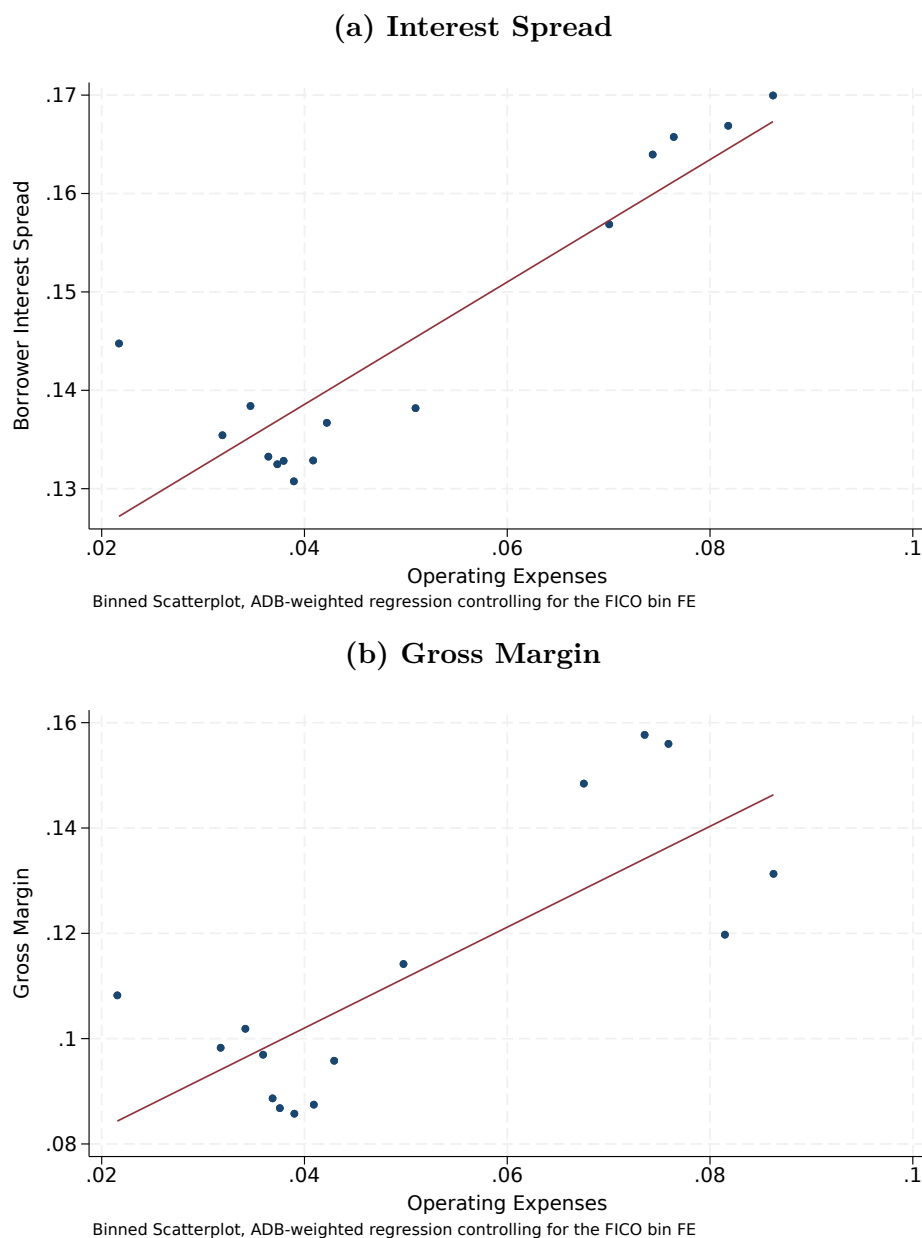


Figure 5: Return on Assets (ROA) for Borrowers

This figure plots all components of income and losses/expenses (left y-axis) and the return on assets (ROA) (black line, right y-axis) for borrowers by origination FICO bin. Income components are plotted as positive values, credit losses and expenses appear as negative values. Interest income, interest expense and credit loss are computed as in [Figure 2](#). Non-interest income includes interchange income ([Figure 3b](#)) plus fee income. Non-interest expense includes rewards expenses ([Figure 3b](#)), operating expenses, interchange expenses and fraud expenses. The variables are reported as average lifetime rates, which are computed by cumulating the lifetime dollar amount of the quantity for all accounts in the FICO bin over the entire sample period, divided by the FICO bin's cumulative Average Daily Balance (ADB). ROA (net margin) equals interest income minus interest expense and net charge-offs, plus net interchange income (interchange minus rewards) and fee income, minus operating expenses and other non-operating expenses. All rates are annualized.

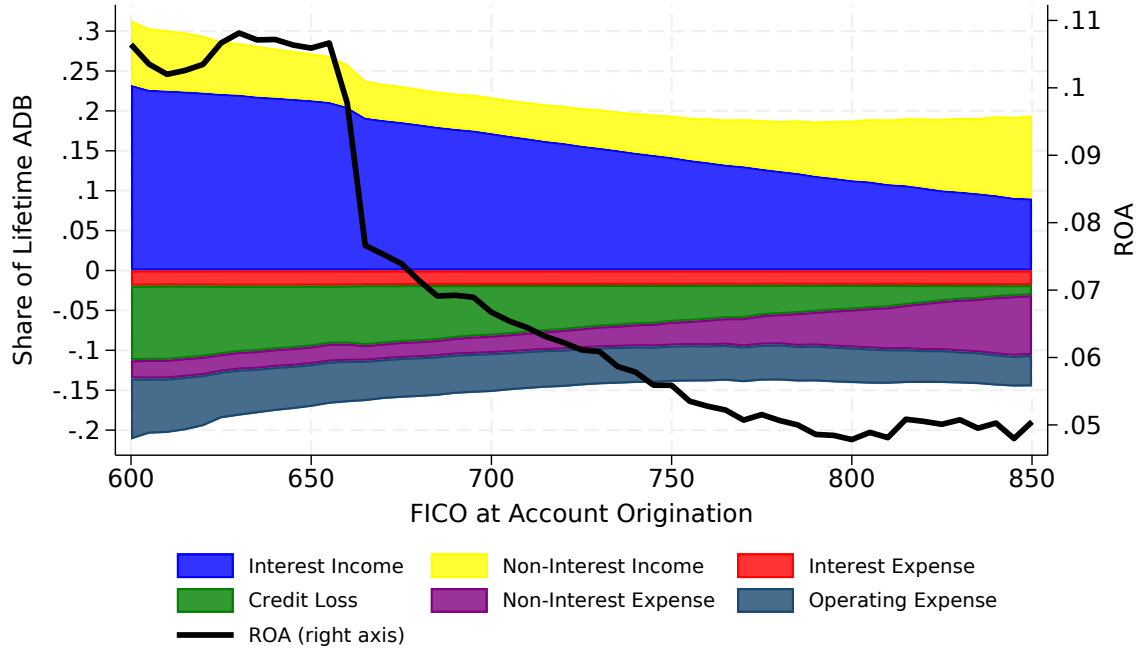
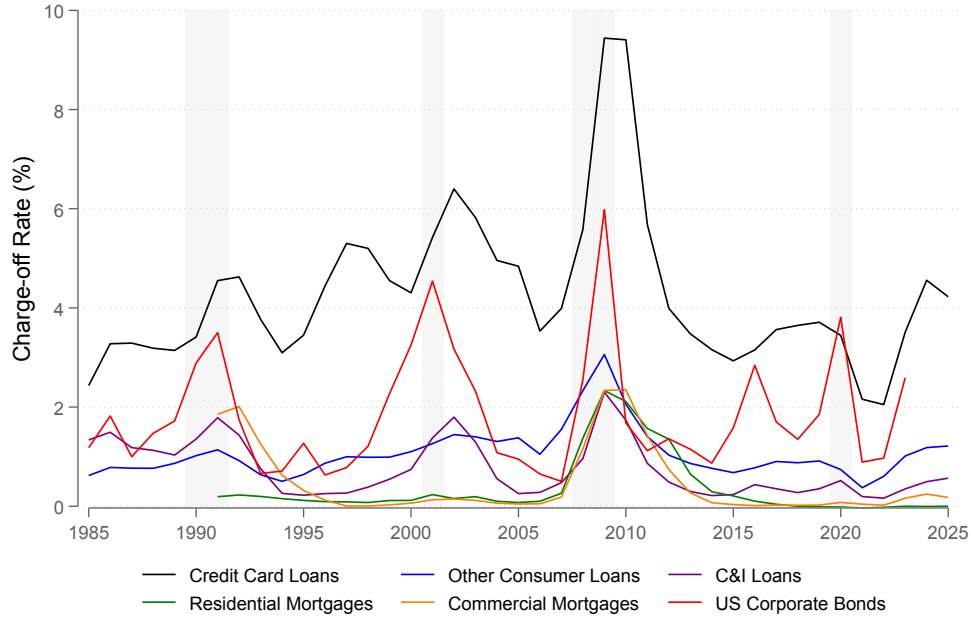


Figure 6: Default rates across various types of loans and corporate bonds

This figure presents the time series of charge-off/default rates for several types of loans and corporate bonds. Panel (a) displays the net charge-off rates for credit cards, other consumer loans, commercial and industrial (C&I) loans, single-family residential mortgages, and commercial real estate loans, obtained from FRED. The U.S. corporate bond default rate is obtained from Standard & Poor's (S&P), which reports the number of issuers that defaulted in a given period divided by the total number of issuers at the beginning of that period. Panel (b) compares the default rate of U.S. speculative-grade corporate bonds from S&P with the credit card charge-off rate.

(a) Bank loans and corporate bonds



(b) Credit cards and speculative grade corporate bonds

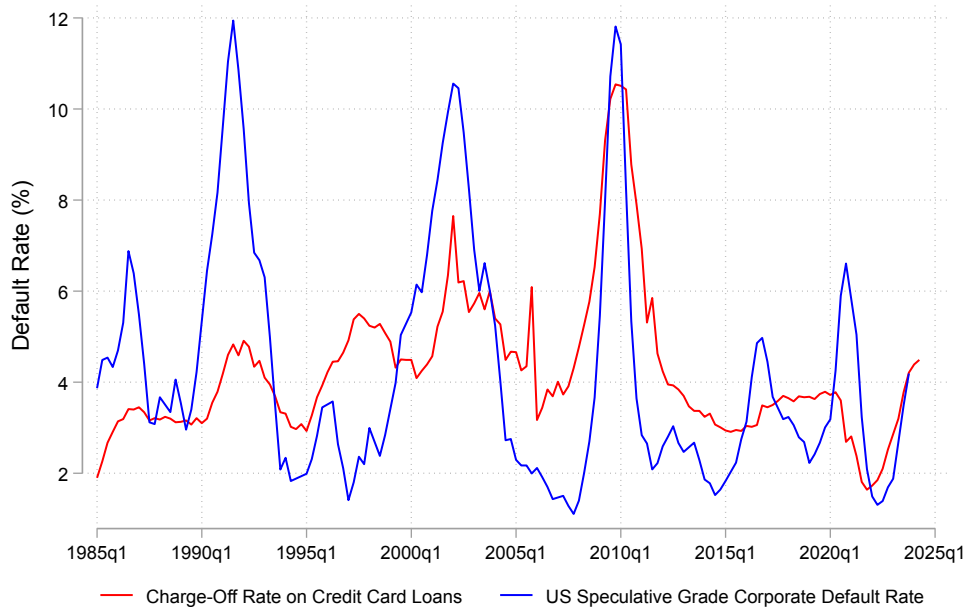


Figure 7: Credit card charge-off dynamics: account life cycle and time series trends

This figure plots the charge-off rates for borrowers of selected origination FICO scores. Panel (a) plots the charge-off rates against account age (months since origination) for accounts originated in 2016, for origination FICO scores of 600 (blue), 660 (red), 700 (green), 760 (yellow), and 800 (purple). Charge-off rate is calculated as total borrower charge-offs divided by total borrower Average Daily Balance (ADB) at the given age. Panel (b) presents the time series of charge-off rates for the origination FICO bins. The charge-off rate is the total borrower charge-offs divided by total borrower ADB for the given bin in the given month. All rates are annualized.

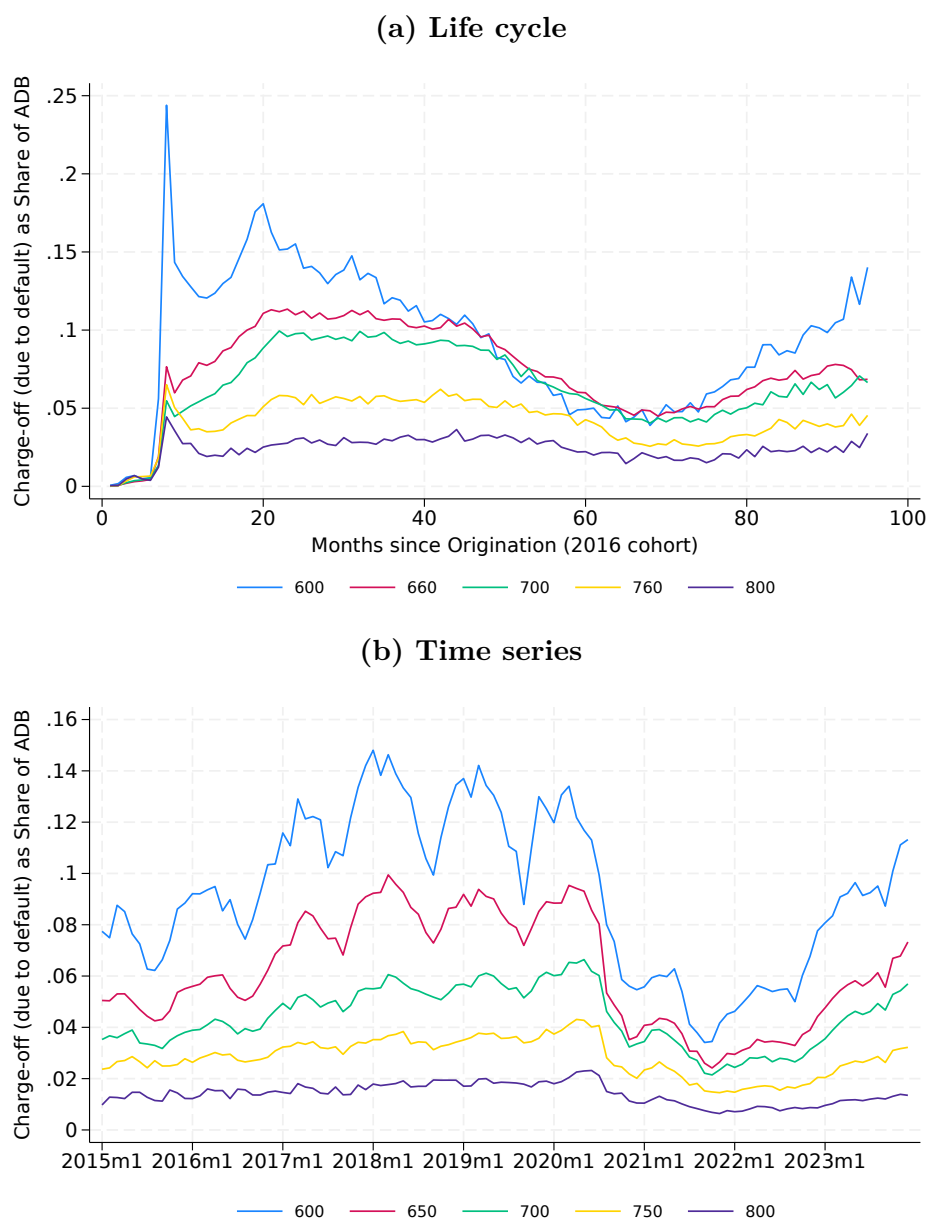


Figure 8: Risk exposure by origination FICO score

This figure plots the average charge-off rate (blue line, left y-axis) and estimated charge-off beta (black line, right y-axis) for each origination FICO bin. We estimate charge-off beta by regressing the change in the FICO bin's quarterly charge-off rate on the change in the charge-off rate of the aggregate credit card portfolio as in equation (1).

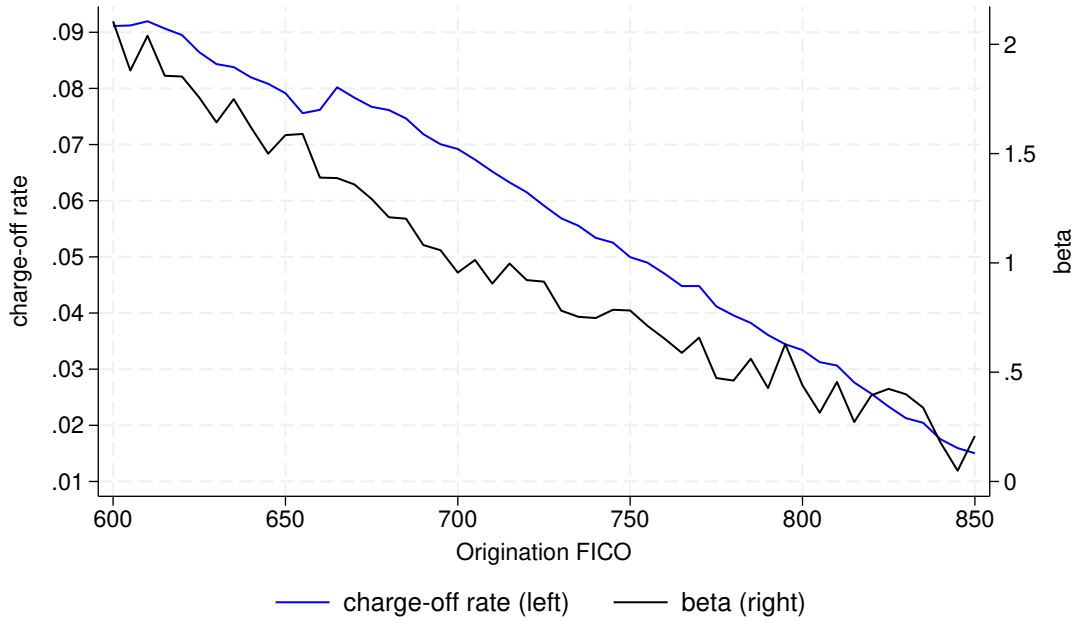


Figure 9: Credit card ROAs and risk premiums vs. corporate bond spreads

This figure compares the average Return on Assets (ROA) for borrowers (blue line), with the fitted ROA (black line) obtained by estimating equation (2), and the ROA of transactors (orange dot), who are default-free by construction. The figure also compares the estimated risk premium component (purple line) to the estimated risk premium on corporate bonds (default-adjusted yield spread, green line). The default-adjusted spread is estimated using the buy-and-hold-to-maturity returns on portfolios of corporate bonds formed on the basis of their initial credit rating and month of issuance. In the event that a bond defaults, we assume a 40% recovery rate, per Standard & Poor's (S&P) estimates. The buy-and-hold returns are adjusted for the risk-free rate by subtracting the yield on a Treasury with the same maturity on the bond's issuance date. The average of the adjusted returns is our estimate of the default-adjusted credit spread for a given bond rating. We map bond ratings to FICO scores based on expected 5-year default rates, as estimated by S&P for bonds and our calculations for credit cards. Shaded areas represent 95% confidence intervals.

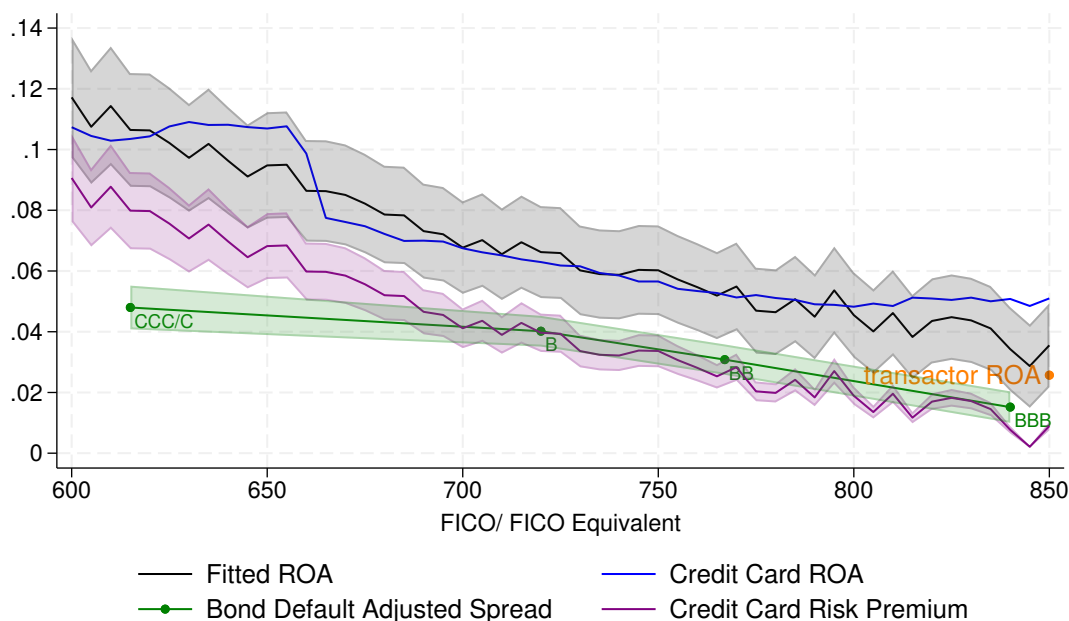


Table 1
Summary statistics

Panel A presents summary statistics for components of income and expenses at the bank-month level, based on Y-14 portfolio-level data from January 2015 until December 2023. The analysis is restricted to general-purpose consumer credit cards. All variables are scaled by each bank’s month-end balance and annualized. Panel B presents summary statistics for account-level balance, fee, and rate variables for accounts in the cleaned sample that were originated in January 2015. Moments for fees, non-interest charge, charge-offs and recoveries are computed only for observations where the corresponding variable has a non-zero value. A borrower is defined as an account that either revolves a balance (does not repay the full balance within the grace period) or is charged off in a given month. Active transactors are accounts in a given month that do not revolve a balance (are not borrowers) and have either a positive cycle-ending balance, purchase volume, or payment activity in that month.

Panel A: Portfolio Level					
	mean	s.d.		mean	s.d.
Interest			Loss		
Interest Income	0.142	0.125	Charge-off	0.050	0.023
Interest Expense	0.024	0.022	Recovery	0.005	0.004
Non-Interest Income			Non-Interest Expense		
Fee Income	0.017	0.015	Reward	0.048	0.030
Interchange	0.071	0.056	Collection	0.020	0.010
Other	0.004	0.008	Fraud	0.003	0.002
			Other Operating	0.049	0.078

Panel B: Account Level				
	mean	p25	p50	p75
Borrowers				
ADB (\$)	3340.33	805.59	2058.78	4382.68
Purchase Volume (\$)	507.28	0.00	50.72	392.82
Finance Charge (\$)	45.43	4.03	24.49	62.32
Effective Interest Rate (%)	16.63	12.30	20.10	23.60
APR (%)	17.47	14.99	20.24	23.49
Credit Limit (\$)	7353.82	2300.00	5000.00	10000.00
Late Fee (\$)	28.39	25.00	35.00	37.00
Annual Fee (\$)	67.10	39.00	49.01	95.00
Balance Transfer Fee (\$)	180.00	46.02	97.25	180
Total Non-interest Charge (\$)	37.25	25.00	35.00	39.00
Charge-off (\$)	3721.34	941.99	2391.02	4843.18
Recovery (\$)	1424.25	187.72	612.29	1750.00
Active Transactors				
ADB (\$)	750.32	4.86	154.73	726.97
Purchase Volume (\$)	1312.33	58.00	411.16	1437.76
APR (%)	16.35	14.24	16.99	20.99
Credit Limit (\$)	10895.47	4600.00	9000.00	15000.00
Annual Fee (\$)	80.29	39.00	75.00	95.00
Total Non-interest Charge (\$)	49.32	2.91	39.00	94.68

Table 2
Profit components by account type and origination FICO score

This table presents the components of profits by origination FICO score bin for borrowers, transactors, and all accounts, respectively. The bins correspond to 20-point increments and are for accounts originated between January 2015 and December 2017. ADB share is the proportion of lifetime Average Daily Balance (ADB) of the accounts in the FICO bin out of the total ADB of accounts in that panel. All average rate variables are constructed using the same methodology: for each origination FICO bin, we cumulate the monthly dollar value of the variable of interest across all accounts in that bin over their lifetime, then divide by the accounts' cumulative lifetime ADB. Variable definitions are provided in Section 4. A borrower is defined as an account that either revolves a balance (does not repay the full balance within the grace period) or is charged off in a given month. Active transactors are accounts that do not revolve a balance (are not borrowers) but have either a positive cycle-ending balance, purchase volume, or payment activity in that month. All rates are annualized.

Origination FICO	ADB Share (%)	Interest Spread (%)	Net Charge-off (%)	Interchange (%)	Reward (%)	Fee (%)	Operating Expense (%)	ROA (%)
Panel A: Borrowers								
600	1.30	21.28	9.35	2.91	2.15	5.13	7.52	10.64
620	2.71	20.44	8.92	2.69	2.15	4.36	6.35	10.41
640	5.34	19.84	8.06	2.53	2.09	3.68	5.55	10.71
660	8.89	18.45	7.43	2.39	2.00	2.87	5.19	9.35
680	14.38	16.62	7.09	2.28	1.90	2.17	4.97	7.24
700	14.62	15.55	6.41	2.45	2.07	1.99	4.86	6.76
720	13.48	14.32	5.72	2.69	2.31	1.92	4.70	6.28
740	11.66	13.14	5.04	3.01	2.63	1.89	4.60	5.84
760	9.43	11.96	4.43	3.46	3.05	1.91	4.53	5.37
780	7.17	10.86	3.82	4.17	3.69	2.01	4.50	5.08
800	5.31	9.72	3.17	5.10	4.54	2.16	4.46	4.84
820	3.43	8.74	2.45	6.01	5.59	2.44	4.21	4.98
840	1.90	7.77	1.65	6.90	6.81	2.68	3.95	4.98
850	0.39	7.22	1.27	7.55	7.40	2.86	3.95	5.05
Average (ADB-weighted)		14.55	5.75	3.12	2.72	2.31	4.84	6.79
Panel B: Transactors								
600	0.3	-	-	54.56	41.45	14.09	6.91	16.43
620	0.71	-	-	50.59	41.66	10.29	5.62	9.22
640	1.46	-	-	49.14	41.68	8.55	4.85	6.68
660	2.57	-	-	46.99	40.10	6.62	4.58	4.39
680	4.35	-	-	44.56	37.77	5.16	4.50	2.89
700	5.79	-	-	43.29	37.11	4.89	4.38	2.70
720	7.29	-	-	42.19	36.68	5.15	4.25	2.89
740	8.88	-	-	40.96	36.24	5.29	4.12	2.77
760	10.39	-	-	39.93	35.30	5.24	4.17	2.87
780	12.74	-	-	39.12	34.06	4.84	4.32	3.01
800	16.32	-	-	38.37	32.66	4.33	4.48	3.21
820	15.48	-	-	35.39	31.61	4.48	3.99	2.01
840	10.9	-	-	31.53	30.57	4.7	3.57	0.16
850	2.83	-	-	31.41	30.04	4.68	3.55	0.29
Average (ADB-weighted)	-	-	-	38.8	34.15	4.97	4.2	2.57
Panel C: All Users								
600	1.18	20.48	9.03	4.66	3.46	5.43	7.50	10.82
620	2.45	19.59	8.59	4.52	3.61	4.58	6.32	10.37
640	4.84	18.96	7.74	4.41	3.63	3.87	5.53	10.55
660	8.08	17.58	7.13	4.30	3.56	3.02	5.17	9.14
680	13.09	15.79	6.79	4.16	3.44	2.30	4.95	7.05
700	13.48	14.55	6.05	4.80	4.01	2.15	4.83	6.53
720	12.69	13.09	5.30	5.70	4.85	2.16	4.67	6.02
740	11.30	11.60	4.53	6.95	6.03	2.24	4.55	5.53
760	9.55	10.00	3.81	8.67	7.56	2.38	4.48	5.02
780	7.88	8.19	3.03	11.55	10.00	2.59	4.46	4.64
800	6.73	6.08	2.18	15.61	13.31	2.83	4.47	4.33
820	4.98	4.48	1.47	17.89	15.98	3.25	4.12	3.78
840	3.06	3.32	0.89	18.32	17.70	3.61	3.78	2.60
850	0.70	2.48	0.61	20.03	19.13	3.80	3.75	2.56
Average (ADB-weighted)		12.41	5.01	7.81	6.76	2.65	4.76	6.24

Table 3
Credit card profit components and bank operating expenses

This table presents regressions of borrower interest spread, net charge-off rate, and gross margin on the operating expense rate. Observations are aggregated to the bank-origination-FICO bin level from account-level data. Interest spread is defined as the sum of finance charges minus interest expense for all borrower observations in the bin, divided by the borrowers' total Average Daily Balance (ADB). Similarly, the net charge-off rate is the total charge-offs minus recoveries divided by borrowers' total ADB. Gross margin is computed as total gross profit over the borrowers' total ADB. Operating expense rate is the total operating expense divided by the total cycle-ending balance, measured at the bank-month level and averaged over the sample period from January 2015 to December 2023. The regressions are weighted by the ADB of the FICO bin. Standard errors in parentheses are clustered at the bank level. ***, **, and * denote significance at the 1%, 5%, and 10% (two-sided) levels, respectively.

	Interest Spread	Net Charge-off	Gross Margin
	(1)	(2)	(3)
Operating Expense Rate	0.622*** (0.119)	0.139 (0.149)	1.127** (0.528)
Constant	0.114*** (0.008)	0.043*** (0.007)	0.055** (0.020)
Origination FICO FE	Y	Y	Y
Observations	919	908	908
R^2	0.895	0.734	0.489

Table 4
Risk premium estimates by origination FICO score

This table presents the risk exposure estimate for each origination FICO score bin portfolio based on the two-stage procedure equivalent to [Fama and MacBeth \(1973\)](#). We first estimate a single-factor model of default risk on the cross-section of credit card portfolios. To proxy for the systematic component of default risk, we use the quarterly change in the aggregate credit card portfolio’s charge-off rate. For each origination FICO portfolio i , we regress the change in its quarterly charge-off rate on this proxy (Equation 1) to obtain the estimate of its charge-off beta β_i . We then regress the panel of ROAs of the FICO portfolios on their risk exposures β_i to estimate the risk premium γ for exposure to default risk and the zero-beta rate λ (Equation 2). We value weight the ROA observations based on the portfolio’s ADB. Standard errors, reported in parentheses, are clustered at the quarter and portfolio level.

Parameter	γ	λ			
Estimates	0.0430 (0.0035)	0.0266 (0.0070)			
Origination FICO	β	ROA	Fitted ROA	Risk Premium	Risk-adj. ROA
600	2.106 (0.342)	10.74%	11.72% (1.02%)	9.06% (0.74%)	1.67% (0.74%)
620	1.853 (0.215)	10.44%	10.63% (0.95%)	7.97% (0.65%)	2.46% (0.65%)
640	1.620 (0.136)	10.82%	9.63% (0.90%)	6.97% (0.57%)	3.84% (0.57%)
660	1.390 (0.078)	9.87%	8.64% (0.85%)	5.98% (0.49%)	3.89% (0.49%)
680	1.209 (0.061)	7.22%	7.86% (0.82%)	5.20% (0.42%)	2.02% (0.42%)
700	0.956 (0.048)	6.75%	6.77% (0.78%)	4.11% (0.33%)	2.64% (0.33%)
720	0.921 (0.063)	6.29%	6.62% (0.77%)	3.97% (0.32%)	2.33% (0.32%)
740	0.748 (0.095)	5.85%	5.87% (0.75%)	3.22% (0.26%)	2.63% (0.26%)
760	0.653 (0.084)	5.34%	5.47% (0.74%)	2.81% (0.23%)	2.53% (0.23%)
780	0.462 (0.099)	5.11%	4.64% (0.72%)	1.99% (0.16%)	3.12% (0.16%)
800	0.440 (0.098)	4.82%	4.55% (0.72%)	1.89% (0.15%)	2.93% (0.15%)
820	0.395 (0.086)	5.09%	4.36% (0.72%)	1.70% (0.14%)	3.40% (0.14%)
840	0.179 (0.115)	5.08%	3.43% (0.70%)	0.77% (0.06%)	4.31% (0.06%)
850	0.208 (0.055)	5.10%	3.55% (0.71%)	0.90% (0.07%)	4.20% (0.07%)